What is a force, really?

? The first law is recited as "a body keeps moving unless pushed". Why is it a definition of the frame rather than a law about forces?

1 A frame you are allowed to use

An inertial frame is one in which a body with no net force keeps a constant velocity. That is a test you can run with a puck on a level air table, and it is also a definition: the first law selects the frames in which F = ma may be written.

So the law is not "motion needs a push". It is "no push means no change", and only in the right room.

2 The same ball in two books

In the ground frame the ball on a table has zero net force and zero acceleration. In the frame of a braking car the same ball drifts forward although nothing touched it: the frame invented a force.

Add the pseudo force $-ma_0$ and the bookkeeping is repaired, which is what the last section of this part does properly.

→ a frame that needs a fix-up force to explain free motion is not inertial

3 How to detect one without looking out

A plumb line hangs off vertical, a water surface tilts, a pendulum swings with a changed period. No sealed box can distinguish "I am accelerating" from "a uniform gravitational field appeared" — that is the equivalence principle speaking.

$$a_{\text{frame}} \neq 0 \;\Rightarrow\; \text{pseudoforcesappearineveryequation}$$

4 A worked test with real numbers

A puck slides on a level air table at 3 m/s while the room is at rest. A second puck is released inside a van braking at 2 m/s². What does each observer measure?

$$\text{ground}: \; a = 0 \quad \text{van}: \; a = 2 \text{ m/s}^2 \text{ forward}$$

→ the van observer must invent f = ma = 2m forward; the ground observer invents nothing

DEFINE

inertial frame

A frame in which a force-free body has constant velocity, and in which F = ma holds without correction terms.

TRAP

the first law is not F = 0 at rest

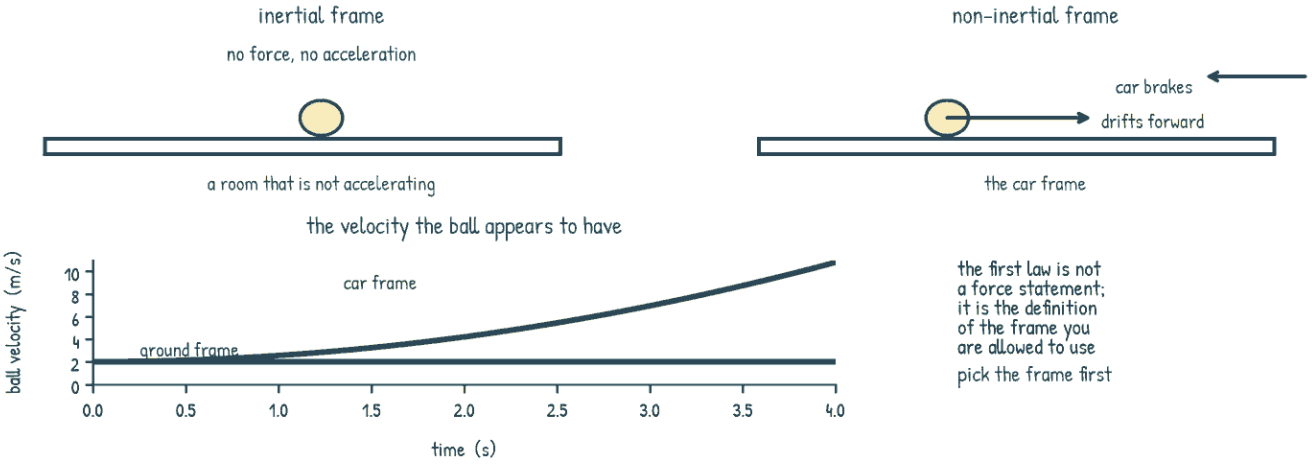
A body moving uniformly has zero net force. Rest is not privileged; constant velocity is.

CHECK

one-line test

If you had to invent a force to explain a motion, you are in a non-inertial frame or you forgot an interaction.

Pick the frame first: in an inertial frame a force-free body keeps its velocity



the first law is not a force statement; it is the definition of the frame you are allowed to use

pick the frame first

KEEP / pick the frame first: a force is what an inertial frame says it is

Is F = ma the law, or a special case?

? Rockets, falling chains and conveyor belts all have mass entering or leaving. Which form of Newton's second law survives all three?

1 The statement that always holds

Force is the rate of change of momentum. For a fixed mass this collapses to ma, which is why the textbook form works almost everywhere — and why it fails exactly where mass is changing.

$$\vec{F} = \frac{d\vec{p}}{dt} = m\vec{a} + \vec{v}\frac{dm}{dt}$$

2 A hit is an area, not a number

In a collision the force is a pulse lasting a millisecond or less. What survives is its integral: the impulse, equal to the momentum change. The area under an F–t graph is the whole story.

$$\vec{J} = \int_{t_1}^{t_2} \vec{F}\,dt = \Delta\vec{p}$$

→ if the pulse is 1000 N for 1 ms, J = 1 N s whether or not you know its shape

3 When mass arrives already moving

Sand dropped on a belt has zero horizontal momentum; the belt must give it v. The force is $\dot{m}v$, not ma, because dm/dt is doing the work.

$$F = \frac{dp}{dt} = \dot{m}\,v \qquad \text{for mass arriving from rest}$$

→ the same term appears as λv^2 when a chain is picked up off the floor

4 A worked impulse budget

A 0.145 kg cricket ball arrives at 30 m/s and leaves at 40 m/s along the same line, contact lasting 1.2 ms. Find the impulse and the mean force.

$$J = m\Delta v = 0.145(40 + 30) = 10.2 \text{ N s}$$

$$\bar{F} = \frac{J}{\Delta t} = \frac{10.2}{1.2 \times 10^{-3}} = 8.5 \text{ kN}$$

→ the ball's weight, 1.4 N, is 0.017 % of that: gravity can be dropped during the hit

KEY RESULT

three forms

$$\vec{F} = m\vec{a}$$

$$\vec{F} = \frac{d\vec{p}}{dt}$$

$$\vec{J} = \Delta\vec{p}$$

Use the second whenever mass changes; the first is a corollary.

TRAP

which mass

In F = ma, m is the mass being accelerated right now. For a rocket it is m(t), and the exhaust momentum must be counted separately.

NUMBERS

feel the scale

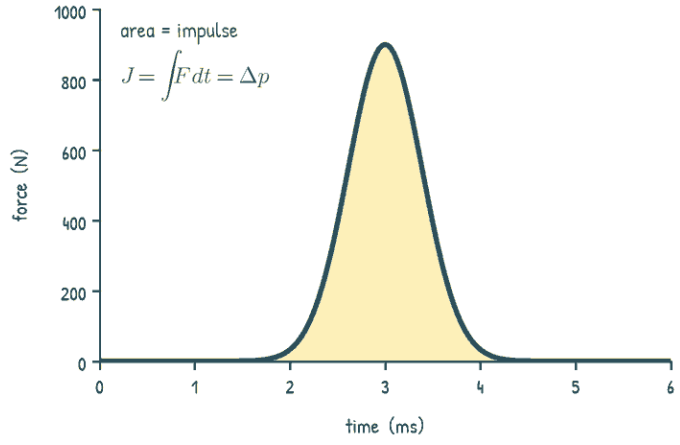
1000 N for 1 ms = 1 N·s

a 0.145 kg ball reversed at 7 m/s needs 2 N·s

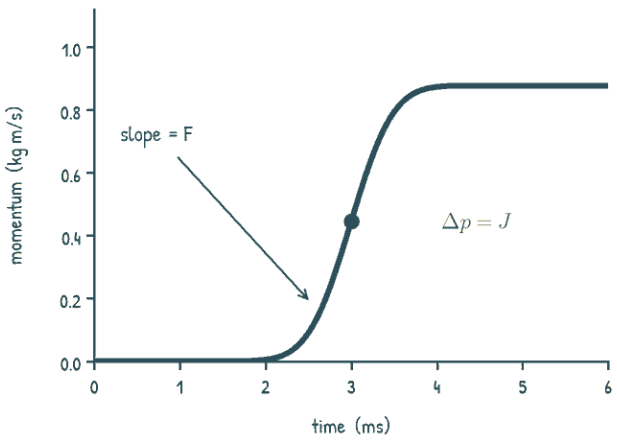
$\dot{m}$ = 2 kg/s at 3 m/s needs 6 N

F = dp/dt is the general law; F = ma is the fixed-mass shortcut, and the impulse is the area

a real hit is a pulse, not a number



the curve of momentum is what F shapes



KEEP / F = dp/dt is the law; F = ma is the fixed-mass shorthand

Why is the normal force not the reaction to weight?

? Equal and opposite sounds like a pair. What extra conditions turn two forces into a third-law pair?

1

Four conditions, not two

A third-law pair is equal in magnitude, opposite in direction, of the same type, and acts on two different bodies. It comes from one interaction, so it dies when the interaction dies.

Push hard on a wall and the wall pushes back just as hard — until you let go, when both forces vanish together.

2

The block on the table

The table pushes the block up with N. The block pushes the table down with an equal N. That is the pair. The block's weight pairs with the block pulling the Earth up, a force nobody draws.

$$N_{\text{onblock}} \;\leftrightarrow\; N_{\text{ontable}}$$

$$mg_{\text{onblock}} \;\leftrightarrow\; mg_{\text{onEarth}}$$

→ N and mg act on the same body and are different in type: never a pair

3

Why they cancel only sometimes

On a table at rest, N = mg and the two cancel, which hides their different origins. In a lift accelerating upward N > mg, and in free fall N = 0 while mg is unchanged.

$$N - mg = ma \;\Rightarrow\; N = m(g + a)$$

4

A worked pair audit

A 2 kg block rests on a table. List the forces on the block and on the table, then pair them.

$$\text{block}: \; N = 19.6\ \text{N}\uparrow,\; mg = 19.6\ \text{N}\downarrow$$

$$\text{table}: \; 19.6\ \text{N}\downarrow \text{ from block},\; 19.6\ \text{N}\uparrow \text{ from floor}$$

→ the third-law partner of N on the block is the 19.6 N push on the table, not mg

DEFINE

third-law pair

Same type, equal size, opposite direction, different bodies, one interaction.

TRAP

the popular wrong pair

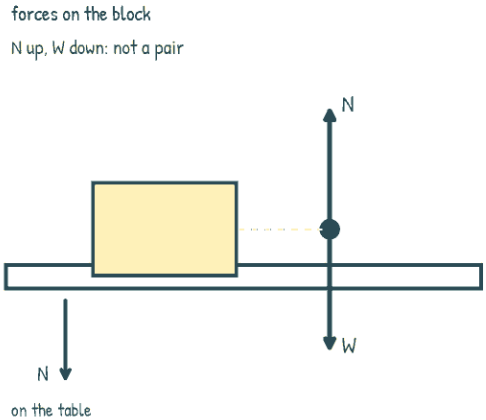
N and mg are not a pair: same body, different interaction. They balance only when a = 0.

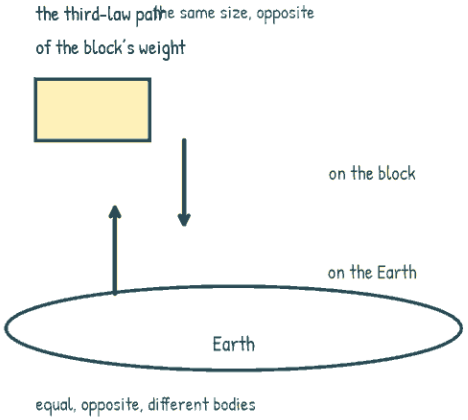
CHECK

name both bodies

Say the sentence out loud: "the table pushes the block, the block pushes the table." If the two names are not different, it is not a pair.

Equal and opposite, but never on the same body: the normal force is a contact pair, not a weight pair





KEEP / a pair needs two bodies and one interaction; N and mg are neither

How do you draw a diagram that never lies?

? Six forces, one acceleration. What procedure makes the equations fall out in one pass?

1

Isolate exactly one body

Draw the body alone. Every contact becomes a normal and possibly a friction; every rope becomes a tension along the rope; every mass gets its weight. Nothing else is allowed in.

If you cannot name the agent that exerts a force, erase the force. "Centrifugal" and "the force of motion" fail this test immediately.

2

Choose axes along the acceleration

On an incline the acceleration is along the slope, so resolve along and perpendicular to it. The two equations then decouple: one gives N, the other gives a.

$$mg\sin\theta - f = ma$$

$$N = mg\cos\theta$$

→ N is less than mg on a slope, equal to mg only on level ground

3

The two errors that survive to the exam

Counting the centripetal force as an extra arrow, and counting the pseudo force without declaring a non-inertial frame. Both add a force that is already the resultant.

→ also: a force on the body you did not isolate belongs somewhere else

4

A worked incline

A 5 kg block slides down a 30° rough incline with μ_k = 0.2. Find N, the friction, and the acceleration.

$$N = mg\cos 30^\circ = 42.4\ \text{N}, \qquad f = \mu_k N = 8.5\ \text{N}$$

$$a = g(\sin 30^\circ - \mu_k \cos 30^\circ) = 9.8(0.5 - 0.173) = 3.2\ \text{m/s}^2$$

→ N is 42.4 N, not the 49 N weight: that difference is the whole point of resolving

GATE

the five steps

1 isolate 2 contacts 3 ropes 4 weight 5 axes along a

Then write ΣF = ma twice, on the two axes.

TRAP

double counting

Centripetal force is the name of the inward resultant, not a fifth arrow. Add it and you have solved 2mv²/R = 2mv²/R.

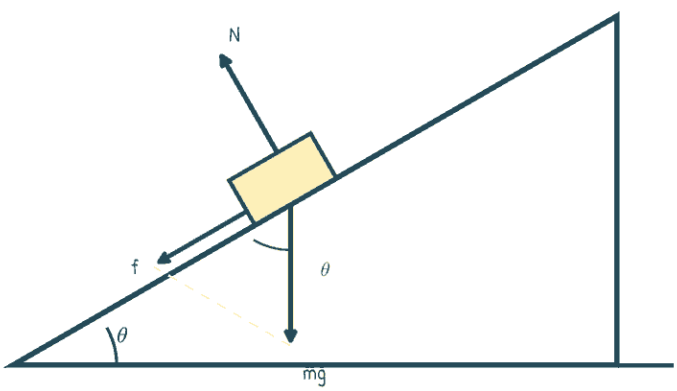
CHECK

sanity pass

Zero acceleration must reproduce the statics answer. Set a = 0 and see.

Isolate the body, draw only external forces, put the axes along the acceleration

isolate, then rotate the axes



$mg\sin\theta - f = ma$

$N = mg\cos\theta$

axes along the acceleration

decouple the two equations

$N \neq mg$ unless $\theta = 0$

KEEP / isolate one body, name every agent, put the axes along the acceleration

What does a weighing scale actually measure?

? In a lift the scale reads more, less, or nothing at all. Which force is it reporting?

1 The four standard forces

Weight mg, always vertically down, always the same mg. Normal N, perpendicular to the surface, can only push. Tension T, along the string, can only pull. Spring force kx, toward the natural length.

None of these is defined by a formula that always holds: N is whatever the constraint requires, up to the point where contact is lost.

2 The scale reads N, not mg

A spring scale measures the force squeezing it, which is the normal force on your feet. Apply Newton to the person:

$$N - mg = ma \;\Rightarrow\; N = m(g + a)$$

$$\text{liftup} : m(g+a) \quad \text{freefall} : 0$$

→ accelerating down at a < g gives a reading of m(g − a), still positive

3 Weightless is not gravity-free

An astronaut in orbit still feels 8.7 m/s² of gravity. Everything around falls with the same acceleration, so nothing presses and N = 0. The scale cannot tell weightlessness from free fall, because they are the same thing.

4 A worked lift ride

A 60 kg passenger stands on a scale in a lift that accelerates up at 2 m/s², then brakes at 2 m/s². What does the scale read?

$$N = m(g + a) = 60(11.8) = 708 \text{ N}$$

$$N = m(g - a) = 60(7.8) = 468 \text{ N}$$

→ true weight is 588 N throughout; only the reading changes

KEY RESULT

reading on a scale

$$N = m(g + a) \;\; (\text{uppositive})$$

up at g/2 reads 1.5 mg; down at g/2 reads 0.5 mg

TRAP

"my weight changed"

mg is fixed by mass and the local field. Only N changes. A scale reads N.

NUMBERS

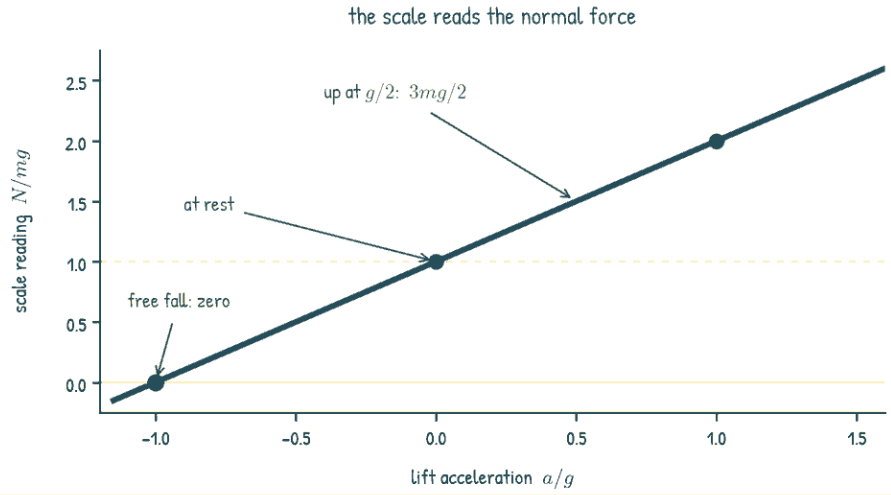
quick numbers

a = g: double weight, zero weight

a = g/3 up: 1.33 mg

spring balance in a lift reads the tension, not mg

Weight never changes; the reading is the normal force, and free fall makes it zero



at rest

$$N = mg$$

accelerating up

$$N = m(g + a)$$

accelerating down

$$N = m(g - a)$$

free fall

$$N = 0$$

the weight never changes

KEEP / weight never changes; the reading is the normal force

Three forces, no acceleration: why does the triangle close?

? A lamp hangs on two wires. When is the sine rule faster than resolving components?

1 Equilibrium is a vector statement

Zero net force means the three arrows, placed head to tail, return to the start. That closed triangle is the whole content of the equilibrium condition; components are just one way to read it.

$$\vec{F}_1 + \vec{F}_2 + \vec{F}_3 = 0$$

2 Lami is the sine rule in disguise

In the force triangle, side F_1 is opposite the angle between F_2 and F_3. The sine rule then gives all three ratios at once.

$$\frac{F_1}{\sin\alpha_1} = \frac{F_2}{\sin\alpha_2} = \frac{F_3}{\sin\alpha_3}$$

→ α_1 is the angle between F_2 and F_3 — the one opposite F_1

3 Where it stops working

Four or more forces: the polygon still closes, but there is no single sine-rule shortcut. A couple or a body with size: force equilibrium alone is not enough, and moments must be added.

4 A worked lamp

A 100 N lamp hangs from two wires making 30° and 60° with the ceiling. Find both tensions by Lami.

$$\frac{T_1}{\sin 120^\circ} = \frac{T_2}{\sin 150^\circ} = \frac{100}{\sin 90^\circ}$$

$$T_1 = 86.6\ \text{N}, \qquad T_2 = 50.0\ \text{N}$$

→ resolving gives the same pair, in twice the lines

KEY RESULT

Lami

$$\frac{F_1}{\sin\alpha_1} = \frac{F_2}{\sin\alpha_2} = \frac{F_3}{\sin\alpha_3}$$

each force over the sine of the angle opposite it

TRAP

which angle

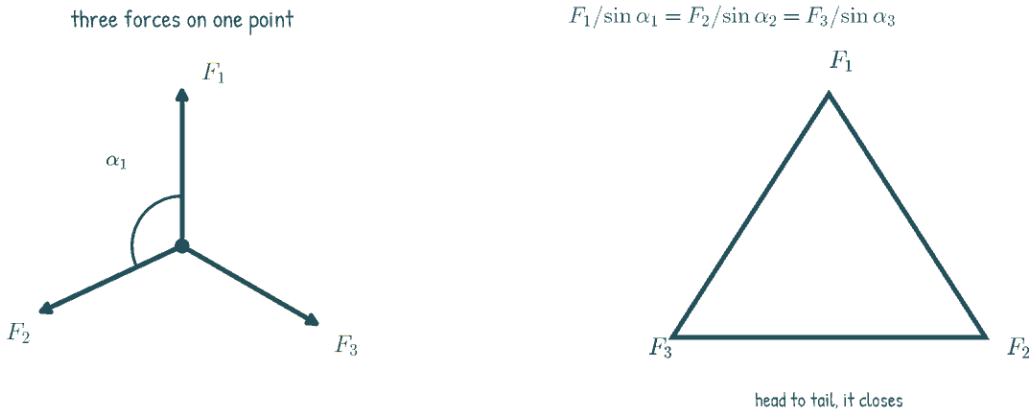
The angle is between the other two forces, not between the force and the horizontal. Using the complement gives a wrong ratio that looks plausible.

GATE

when to reach for it

Exactly three forces, all concurrent, body at rest or moving uniformly. Otherwise resolve.

Three forces, one point, zero sum: the vector triangle obeys the sine rule



KEEP / three forces in equilibrium close a triangle, and the sine rule reads it

Is friction equal to mu N?

? A 20 N block sits on a rough floor while you push with 3 N and it does not move. What is the friction force?

1

Static friction is a range

A block that has not started to slide has zero acceleration, so friction must exactly cancel the push. It takes whatever value is needed, from zero up to a ceiling set by the surfaces.

$$0 \le f_s \le \mu_s N$$

→ push with 3 N and it does not move: f_s = 3 N, not μN

2

At the ceiling it lets go

Increase the push and f rises with it, one for one, until f reaches μ_s N. That instant is the threshold of slipping, and the angle at which a ramp reaches it is the angle of repose.

$$\mu_s = \tan\theta_{\text{repose}}$$

3

Kinetic friction is a value

Once sliding, the force drops to a fixed μ_k N, usually a little below the static maximum. That drop is why a pushed wardrobe suddenly lurches, and why a car's ABS tries to stay just under the threshold.

$$f_k = \mu_k N \qquad \mu_k < \mu_s \text{ usually}$$

4

A worked push

A 4 kg crate sits on a floor with μ_s = 0.5. You push with 15 N, then with 25 N. What is the friction each time?

$$\mu_s N = 0.5(39.2) = 19.6 \text{ N}$$

$$F = 15 \text{ N}: \ f = 15 \text{ N} \qquad F = 25 \text{ N}: \ \text{it slides}$$

→ at 25 N the friction drops to μ_k N, so the crate accelerates rather than creeping

KEY RESULT

the three statements

$$f_s \le \mu_s N, \quad f_s = \mu_s N \text{ at slip}$$

$$f_k = \mu_k N$$

$$\mu_s = \tan\theta_{\text{repose}}$$

TRAP

f = mu N on a stationary body

This is the single most common error in the chapter. On a body that is not slipping, friction is found from equilibrium, then checked against μ_s N.

NUMBERS

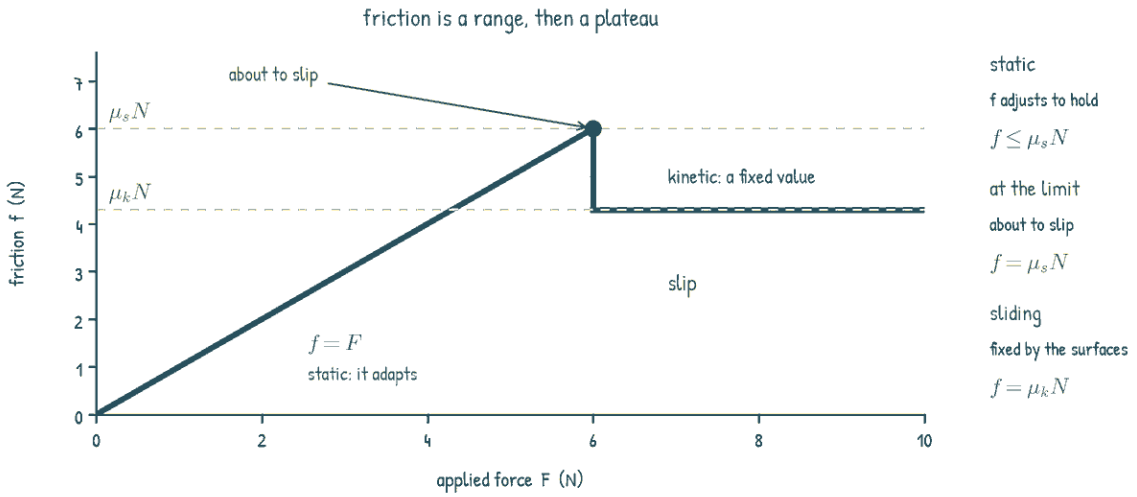
check these

μ_s = 0.6 on 2 kg: ceiling 11.8 N

push 6 N: f = 6 N, still static

push 13 N: it slides with f = μ_k N

Static friction is an inequality with a ceiling; kinetic friction is a value



static

f adjusts to hold

$$f \le \mu_s N$$

at the limit

about to slip

$$f = \mu_s N$$

sliding

fixed by the surfaces

$$f = \mu_k N$$

KEEP / static friction is an inequality with a ceiling, not a number

Which way does friction point?

? A block is dropped on a moving belt and another slides up a ramp. Both are moving, yet friction points in opposite senses. Why?

1 ## Compare the two surfaces, not the body

Friction acts at the interface. What matters is the velocity of the block relative to the surface it touches. If the block lags, friction drags it forward; if it leads, friction holds it back.

$$\vec{f} \text{ opposes } \vec{v}_{\text{block}} - \vec{v}_{\text{surface}}$$

2 ## Three standard pictures

Pushed on a floor: the block would slide right, friction points left. Dropped on a belt: the belt runs forward under it, friction points forward and the block speeds up. Sliding up a ramp: friction points down the slope.

→ on the belt the block's speed increases, so friction does positive work on it

3 ## Where this bites

Walk forward: the floor pushes you forward by friction. Accelerate a car: the road pushes the driving wheels forward. Brake: the same friction points backward on the car. In each case compare the tyre contact patch with the road.

4 ## A worked belt

A 2 kg package is dropped on a belt running at 4 m/s, with μ_k = 0.3. How long until it stops slipping, and how far does it travel?

$$a = \mu_k g = 2.94 \text{ m/s}^2, \qquad t = \frac{4}{2.94} = 1.36 \text{ s}$$

$$s_{\text{pkg}} = \tfrac{1}{2}(4)(1.36) = 2.72 \text{ m}, \qquad s_{\text{belt}} = 5.44 \text{ m}$$

→ the belt slides 2.72 m under the package, and that gap is the friction heat

GATE
the one question
What is the velocity of this body relative to the surface it touches? Friction points opposite to that.

TRAP
opposing the motion in the room
On an accelerating belt, on a turning car, on a walker: friction often points along the motion. It always opposes slip.

CHECK
sign of work
Friction can do positive work on one body and equal negative work on the other. The pair together always removes mechanical energy into heat.

Friction is about relative slip at the contact, not about the body's motion in the room

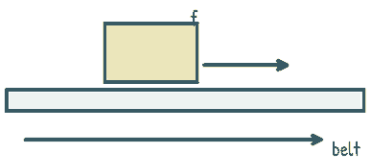
pushed on the floor

it would slide right,
so f points left



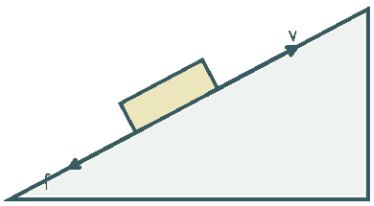
on a moving belt

the block lags behind,
so f points forward



sliding up a ramp

friction acts down
the slope



KEEP / friction opposes relative slip at the contact, not motion in the room

Why is the easiest pull not horizontal?

? Dragging a crate is easier with the rope tilted up. Why, and exactly how much less force is needed?

1 ## The trade

Pulling at angle θ upward reduces the normal force to mg – F sinθ, so it reduces the friction ceiling. But only the horizontal part F cosθ drives the crate. The two effects fight, so there is an optimum.

$$N = mg - F\sin\theta, \qquad f = \mu N$$

$$F\cos\theta = \mu(mg - F\sin\theta)$$

2 ## Minimise the force

Solve for F as a function of θ and minimise. The denominator cosθ + μ sinθ peaks at tanθ = μ, which fixes both the angle and the smallest force.

$$F(\theta) = \frac{\mu mg}{\cos\theta + \mu\sin\theta}$$

$$\theta^* = \tan^{-1}\mu, \qquad F_{\min} = \frac{\mu mg}{\sqrt{1+\mu^2}}$$

→ μ = 0.5 on 2 kg: 26.6° and 8.77 N instead of 9.8 N

3 ## The other use of the same algebra

Let θ grow to 90° and the crate lifts instead of sliding. The angle of repose is the mirror idea: on a ramp, tanθ ≤ μ_s keeps a block still without any external pull.

→ both results come from setting the friction ceiling equal to the driving component

4 ## A worked optimum

A 20 kg crate on μ = 0.4 needs to be dragged. Compare a horizontal pull with the optimum angle.

$$F_0 = \mu mg = 78.4 \text{ N}$$

$$F_{\min} = \frac{78.4}{\sqrt{1+0.16}} = 72.8 \text{ N at } 21.8^\circ$$

→ 5.6 N saved, and the crate unloads 27 N of normal force while being pulled

KEY RESULT

minimum pull

$$\tan\theta = \mu$$

$$F_{\min} = \mu mg/\sqrt{1+\mu^2}$$

the saving is only about 11 % at μ = 0.5, but it is free

NUMBERS

worked

m = 2 kg, μ = 0.5, g = 9.8

horizontal: F = 9.8 N

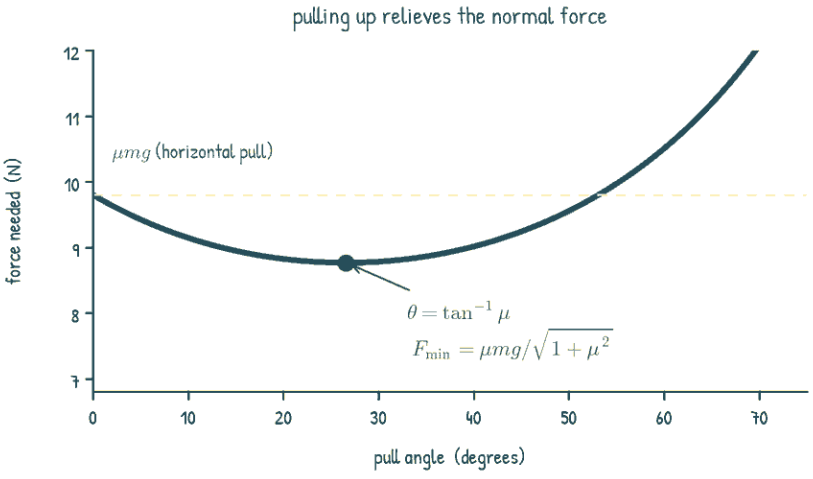
at 26.6°: F = 8.77 N

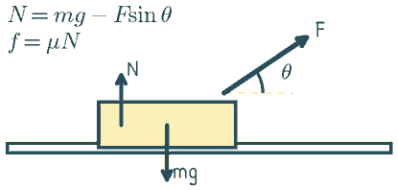
TRAP

two sign errors

N = mg – F sinθ (not plus) when you pull upward. And the friction term uses this reduced N, not mg.

Tilting the pull up steals normal force from the floor, and the best tilt is exactly arctan(mu)





KEEP / tilting up trades normal force for drive, and the best tilt is arctan mu

When do two stacked blocks stop moving together?

? You push the lower block of a stack. What limits how hard you can push before the top one slips?

1 **Look at the top block first**

The only horizontal force on the upper block is friction from the lower one. Its acceleration is therefore f/m, and it can never exceed μ_s g, no matter how hard you push below.

$$a_{\text{top}} = \frac{f}{m} \le \mu_s g$$

2 **While they move together**

Treat the pair as one body: a = F/(M + m). That is consistent with the top block only while the friction needed stays under the ceiling.

$$F \le \mu_s g(M+m)$$

$$\mu_s = 0.3,\ m = 1\ \text{kg},\ M = 2\ \text{kg} \ \Rightarrow\ F_{\max} = 8.82\ \text{N}$$

3 **After the break**

The top block keeps accelerating at μ_s g while the bottom one accelerates faster. The relative acceleration is constant, so the top block slides backward relative to the bottom — it is not "left behind" by inertia, it is dragged at a fixed maximum rate.

4 **A worked stack**

m = 2 kg on M = 3 kg, μ_s = 0.4 between them, floor frictionless. How hard can you push the lower block?

$$F_{\max} = \mu_s g(M+m) = 0.4(9.8)(5) = 19.6\ \text{N}$$

$$a_{\max} = \mu_s g = 3.92\ \text{m/s}^2$$

→ push 19.6 N and the pair moves at 3.92 m/s²; push 25 N and the top one lags

KEY RESULT

the two numbers

$$F_{\max} = \mu_s g(M+m)$$

$$a_{\max} = \mu_s g$$

the ceiling is set by the upper block, never by the push

TRAP

pushing vs pulling

Which block you push changes the answer: the limit above is for a push on the lower block. Repaint the free-body diagram for the other case.

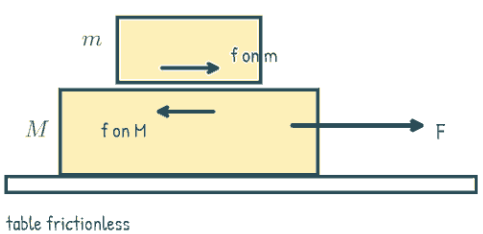
CHECK

limit case

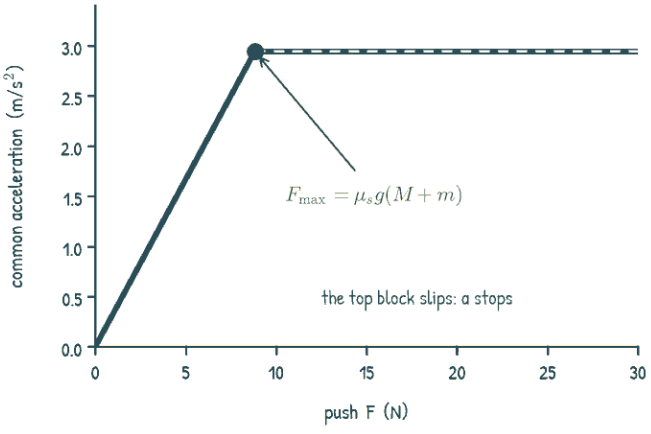
Set μ_s = 0 and the pair cannot move together at all: the top block stays put while the bottom slides out.

The top block is driven by friction alone, so it caps the whole system at a = mu_s g

the friction that drags the top block



together only up to a point



KEEP / the top block is driven by friction alone, so it caps the whole system

System or single body: which do you solve?

? Two blocks are pushed together. How do you get both the common acceleration and the force between them, without solving twice?

1

The system first

Internal forces come in third-law pairs, so they cancel inside a system. A single equation then gives the acceleration of the whole thing.

$$a = \frac{F}{m_1 + m_2}$$

2

Then isolate the far body

Pick the body that has the fewest forces — usually the one you are not pushing. The contact force is its only horizontal force, so it reads off directly.

$$N_{\text{contact}} = m_2 a = \frac{F m_2}{m_1 + m_2}$$

→ equal masses: each carries half. A very light front block carries almost nothing.

3

Why the same trick keeps working

Ropes, wedges and stacked blocks all obey it: sum over the system to get a, then cut the system where the unknown force lives. The cut replaces the interaction with the unknown.

→ with three bodies you may need two cuts, but the recipe is unchanged

4

A worked contact force

F = 30 N pushes m_1 = 2 kg into m_2 = 4 kg on a frictionless floor. Find a and the contact force.

$$a = \frac{30}{6} = 5 \text{ m/s}^2$$

$$N = m_2 a = 20 \text{ N}, \qquad \text{check: } 30 - N = 2(5)$$

→ the check is one line, and it always catches a wrong mass in the denominator

GATE

the recipe

1 system for a 2 isolate for N 3 check with the other body

KEY RESULT

contact force

$$N = \frac{F m_2}{m_1 + m_2}$$

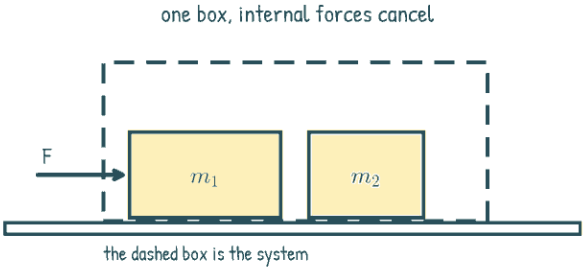
N is not F unless the front block is the whole mass

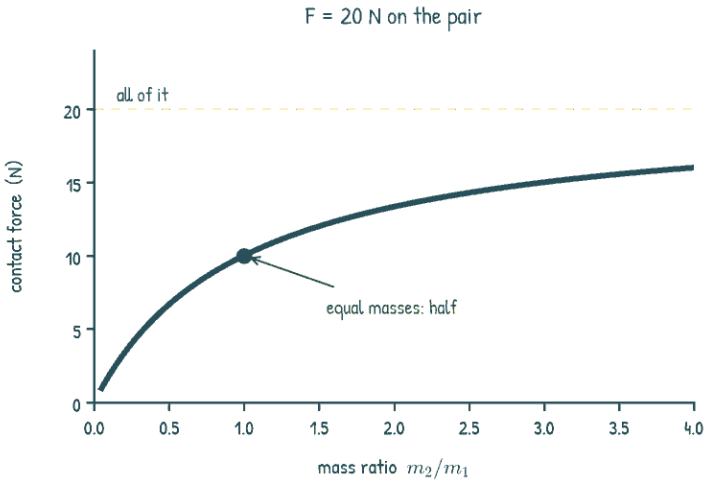
TRAP

adding contact forces to the system

Inside the dashed box the contact forces cancel. If you kept them, you would double count and halve the acceleration.

Use the system for the acceleration, then isolate one body for the force between them





KEEP / use the system for the acceleration, then isolate one body for the internal force

Atwood on a ramp: where does the motion change hands?

? One mass hangs, one sits on a rough 30° ramp. Over what range of mass ratio does nothing move at all?

1 ## The clean Atwood machine

Same tension both sides, same magnitude of acceleration. Adding the two equations eliminates T and gives the standard pair of results.

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2}, \qquad T = \frac{2m_1 m_2 g}{m_1 + m_2}$$

2 ## Put one of them on a ramp

Only the component along the slope drives the ramp mass, so $\sin\theta$ replaces 1 for that side. Add friction and it enters with $\pm$, depending on which way the block is about to slide.

$$a = \frac{(m_2 - m_1\sin\theta)g}{m_1 + m_2} \pm \frac{\mu m_1 g\cos\theta}{m_1 + m_2}$$

3 ## The static band

At rest, friction can supply up to $\mu m_1 g\cos\theta$ in either direction. That gives two thresholds, one for sliding up and one for sliding down, and between them the system is frozen.

$$\sin\theta - \mu\cos\theta \le \frac{m_2}{m_1} \le \sin\theta + \mu\cos\theta$$

→ $\theta = 30°$, $\mu = 0.25$: the band is 0.28 to 0.72

4 ## A worked Atwood

$m_1 = 3$ kg on a smooth 30° ramp, $m_2 = 2$ kg hanging. Find a and T.

$$a = \frac{(2 - 3\sin 30°)9.8}{5} = \frac{0.5(9.8)}{5} = 0.98\ \text{m/s}^2$$

$$T = m_2(g - a) = 2(8.82) = 17.6\ \text{N}$$

→ T is 17.6 N while the hanging weight is 19.6 N: they differ by $m_2 a$

KEY RESULT

the three formulas

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2}$$

$$T = \frac{2m_1 m_2 g}{m_1 + m_2}$$

$$\text{band} : \sin\theta \mp \mu\cos\theta$$

NUMBERS

the worked band

$\theta = 30°$, $\mu_s = 0.25$

0.5 − 0.217 = 0.28 to 0.5 + 0.217 = 0.72

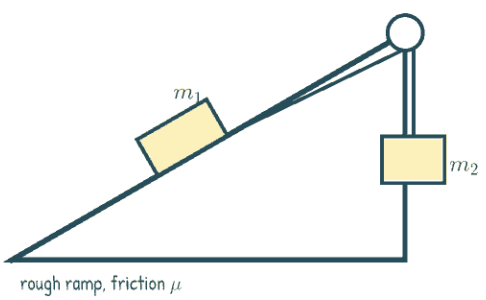
TRAP

T = m g

T equals the hanging weight only when the system is in equilibrium. Otherwise T is found from either body after a is known.

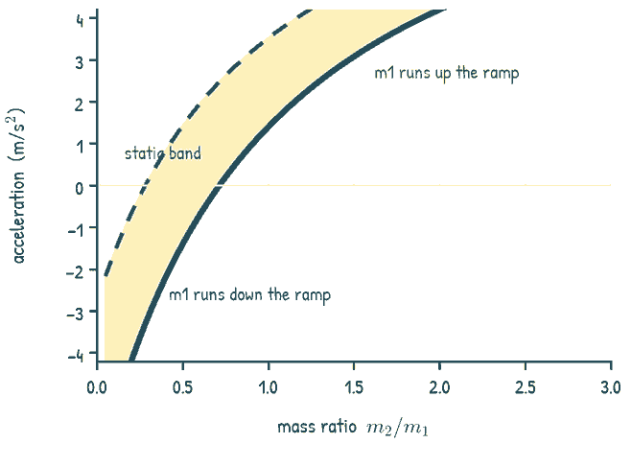
Add friction to the Atwood-on-a-ramp and the acceleration crosses zero at two different mass ratios

one mass on the ramp, one hanging



rough ramp, friction μ

three regimes, one formula



KEEP / friction turns one crossing point into a band of mass ratios

Why does the free end move twice as far?

? A rope runs over a fixed pulley, round a movable one, and is anchored. What links the accelerations, and what links the tensions?

1 ## Write the length, then differentiate

Let x_1 be the segment that ends at the free mass and x_2 the length of each of the two strands holding the movable pulley. The total length is fixed.

$$x_1 + 2x_2 = \text{const} \;\Rightarrow\; a_1 = -2a_2$$

→ the free end must move twice as far, and accelerate twice as hard

2 ## One string, one tension

A massless frictionless pulley cannot support a tension difference, so the rope tension is the same everywhere. The movable pulley therefore feels 2T upward against the weight hanging on it.

$$m_1 g - T = m_1 a_1, \qquad m_2 g - 2T = m_2 a_2$$

3 ## Solve with the constraint

Substitute $a_2 = -a_1/2$ and eliminate T. The free mass always wins the sign argument, because its weight is compared with twice the other weight.

$$a_1 = \frac{2(2m_1 - m_2)g}{4m_1 + m_2}$$

→ equal masses: free end falls at 2g/5, movable pulley rises at g/5

4 ## A worked movable pulley

m_1 = 4 kg at the free end, m_2 = 3 kg on the movable pulley. Find both accelerations and T.

$$a_1 = \frac{2(8-3)9.8}{16+3} = \frac{98}{19} = 5.16\ \text{m/s}^2$$

$$a_2 = -2.58\ \text{m/s}^2, \qquad T = m_1(g - a_1) = 18.6\ \text{N}$$

→ the pulley rises at half the rate the free end falls, and 2T balances m_2 only at rest

KEY RESULT

constraint pair

$$x_1 + 2x_2 = \text{const}$$

$$a_1 = -2a_2, \qquad T \text{ uniform}$$

NUMBERS

equal masses

$m_1 = m_2 = m$

$a_1 = 2g/5$ down, $a_2 = g/5$ up

$T = 3mg/5$

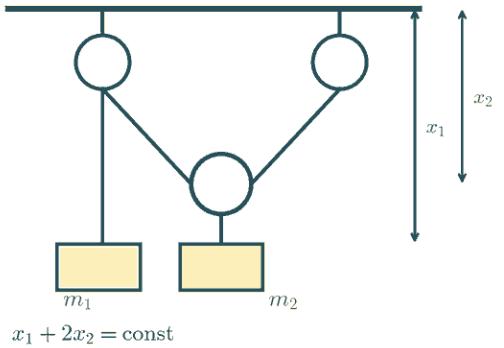
TRAP

sign of the constraint

Decide once which direction is positive for each body. The printed solutions of this chapter are inconsistent about it; the magnitudes 2 : 1 are not.

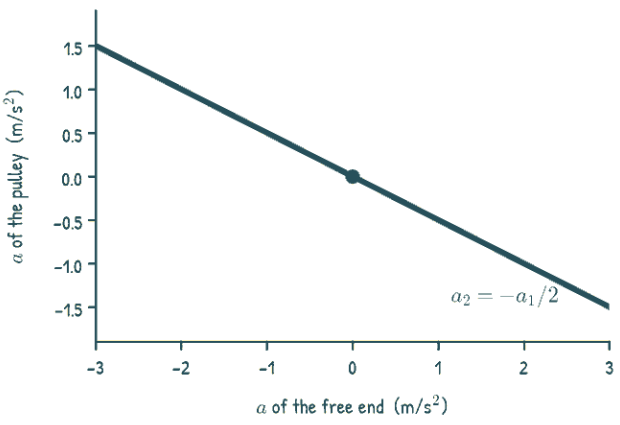
The string length is x1 + 2x2, so the free end accelerates twice as hard as the pulley

one string, three segments



$x_1 + 2x_2 = \text{const}$

differentiate twice



KEEP / the length constraint gives the accelerations; one string gives the tension

Should you ever add a force that is not there?

? In a braking bus a hanging pendulum swings forward. Add a pseudo force, or stay in the ground frame?

1 ### The recipe for an accelerating frame

If the frame accelerates at a_0, every body in it behaves as if an extra force $-ma_0$ acted. Add that one term and Newton's laws work again, with no other changes.

$$\vec{F}_{\text{real}} + (-m\vec{a}_0) = m\vec{a}_{\text{frame}}$$

2 ### The pendulum in the car

In the car frame the bob is at rest with three forces: weight down, pseudo force backward, tension along the string. Resolving gives the hang angle and the tension immediately.

$$\tan\theta = \frac{a_0}{g}, \qquad T = m\sqrt{g^2 + a_0^2}$$

3 ### The ground frame must agree

In the ground frame the bob accelerates forward at a_0. The only real forces are T and mg, and their resultant is horizontal. The same θ falls out — a good check that you have not mixed frames.

$$T\sin\theta = ma_0, \qquad T\cos\theta = mg$$

→ if the two frames disagree, one of them has a force that does not exist

4 ### A worked pendulum in a car

A bob hangs from the roof of a car accelerating at 4.9 m/s². Find the hang angle and the tension per unit mass.

$$\tan\theta = \frac{4.9}{9.8} = 0.5 \;\Rightarrow\; \theta = 26.6^\circ$$

$$T/m = \sqrt{g^2 + a_0^2} = 10.96 \text{ N/kg}$$

→ the same numbers fall out of the ground frame, where T sinθ = ma_0 does the work

KEY RESULT

effective gravity

$$\vec{g}_{\text{eff}} = \vec{g} - \vec{a}_0$$

$$\tan\theta = a_0/g$$

TRAP

mixing frames

The classic disaster: a pseudo force in one equation and ground-frame reasoning in the next. Choose a frame, write it at the top of the page, and stay there.

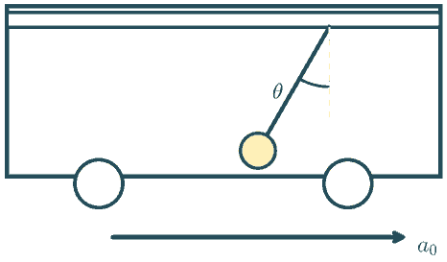
CHECK

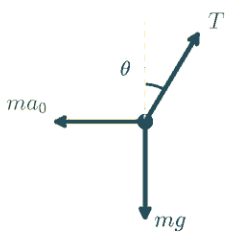
the two-frame test

Solve once each way. Any disagreement is an algebra error, never new physics.

In an accelerating frame, add -ma and then use ordinary equilibrium

accelerating car: the bob hangs askew





$$\tan\theta = a_0/g$$

$$g_{\text{eff}} = \sqrt{g^2 + a_0^2}$$

in the car frame

add the pseudo force

the string lines up with g_eff

KEEP / a pseudo force is frame bookkeeping, not an interaction

Is centripetal force a new force?

? A ball on a string, a car on a bank, a car over a crest. Which real force points inward in each case?

1 ### It is a name, not an interaction

Circular motion at constant speed needs a resultant of mv²/R pointing at the centre. That resultant is produced by whatever real forces happen to point inward. Writing it as an extra force double counts the physics.

$$\sum F_{\text{inward}} = \frac{mv^2}{R} = m\omega^2 R$$

2 ### Three suppliers

A ball on a string: the tension. A car on a bank: the horizontal component of the normal force. A car crossing a hill crest: the difference between weight and normal force, which is why the car feels light.

$$\text{crest}: mg - N = \frac{mv^2}{R}$$

→ at the speed √(gR) the normal force reaches zero and the car leaves the road

3 ### When the supplier quits

A string cannot push, so if the required inward force exceeds the tension available the string goes slack. A tyre cannot pull downward, so on a crest it simply loses contact. The condition is always "the required inward force must be available".

4 ### A worked crest

A car crosses a hill of radius 40 m at 15 m/s. What fraction of its weight does the road still support?

$$N = m\left(g - \frac{v^2}{R}\right) = m(9.8 - 5.63) = 4.17m$$

$$\frac{N}{mg} = 0.43, \qquad v_{\text{lift}} = \sqrt{gR} = 19.8 \text{ m/s}$$

→ at 19.8 m/s the normal force vanishes and the wheels leave the road

DEFINE

centripetal force

The inward component of the resultant, of magnitude mv²/R. It is supplied by real forces; it is not one of them.

TRAP

the extra arrow

Draw mv²/R on the free-body diagram and every equation becomes wrong by that amount. Put it on the other side of the equals sign.

CHECK

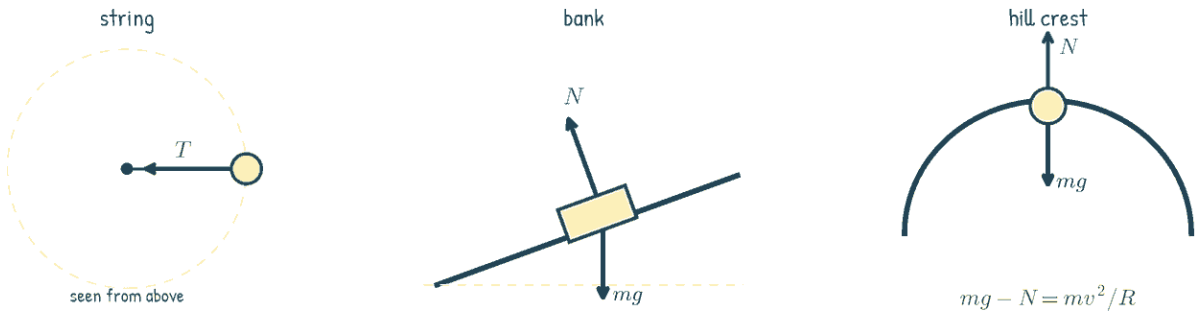
availability test

Compute the inward force required. Compare it with the largest inward force the contact can supply. If required > available, contact is lost.

Centripetal force is the name of the inward resultant, never an extra arrow

$$\sum F_{\text{inward}} = mv^2/R$$

the resultant is named, not added



KEEP / centripetal force is the inward resultant, never an extra arrow

Why are racetrack corners banked?

? With no friction at all, is there still a speed at which a car can take a banked curve?

1 **Resolve vertically and radially**

On a frictionless bank the only forces are weight and the normal, which is tilted. Tilt means it has an inward component, and that component is the centripetal force.

$$N\cos\theta = mg, \qquad N\sin\theta = \frac{mv^2}{R}$$

2 **Divide: the mass disappears**

Dividing the two equations removes both m and N, leaving the design speed. Every vehicle, from a bicycle to a truck, needs the same speed for a frictionless passage.

$$v_0^2 = Rg\tan\theta$$

→ R = 50 m, θ = 22°: v_0 = 14 m/s, about 50 km/h

3 **Why real roads still need friction**

Traffic does not travel at exactly v_0. Below it the car tends to slide down the bank, above it up the bank, and friction has to supply the difference. That is the width of the band on the next sheet.

$$N = \frac{mg}{\cos\theta} > mg$$

4 **A worked bank**

A curve of radius 80 m is banked for 72 km/h with no friction. Find the angle and the normal force on a 1200 kg car.

$$\tan\theta = \frac{v^2}{Rg} = \frac{400}{784} = 0.51 \Rightarrow \theta = 27^\circ$$

$$N = \frac{mg}{\cos\theta} = \frac{11760}{0.891} = 13.2\ \text{kN}$$

→ N exceeds the weight by 12 %, because the bank must also turn the car

KEY RESULT

design speed

$$v_0 = \sqrt{Rg\tan\theta}$$

$$\tan\theta = \frac{v_0^2}{Rg}$$

NUMBERS

typical corner

R = 50 m, θ = 22°, g = 9.8

v_0 = 14.0 m/s = 50.5 km/h

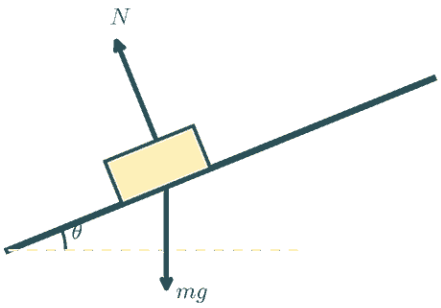
TRAP

N = mg on a bank

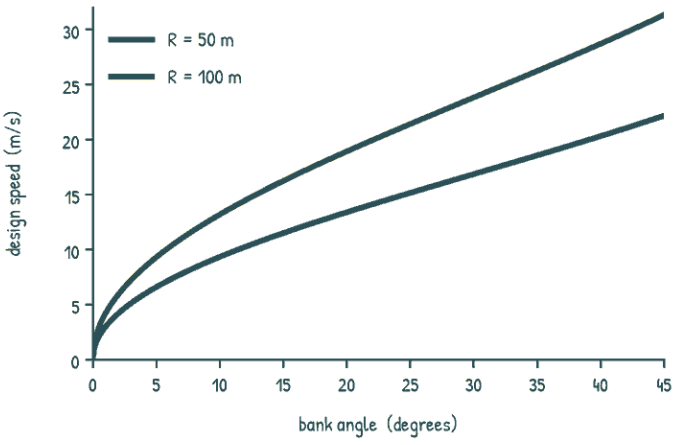
The normal force is larger than mg on a bank: N = mg/cosθ. Using N = mg understates the available friction as well.

With no friction there is exactly one speed that needs no friction at all

$$N\cos\theta = mg$$
$$N\sin\theta = mv^2/R$$
$$v^2 = Rg\tan\theta$$



steeper bank, faster design speed



KEEP / a bank turns normal force into centripetal force without any friction

How much slower or faster can you take the bank?

? Friction turns one design speed into a range. What are the two limits, and what sets them?

1 ## Two limits, one derivation

At the maximum speed the car is about to slide outward, so friction points down the slope. At the minimum it is about to slide inward, so friction points up the slope. Resolve both cases the same way.

$$v_{\max}^2 = \frac{Rg(\sin\theta + \mu\cos\theta)}{\cos\theta - \mu\sin\theta}$$

$$v_{\min}^2 = \frac{Rg(\sin\theta - \mu\cos\theta)}{\cos\theta + \mu\sin\theta}$$

2 ## The three regimes

Below v_min friction points up the slope; at v_0 it is exactly zero; above v_0 it points down. The same formulas cover all three, and v_min is set to zero once μ exceeds tanθ.

→ if μ > tanθ the car can stand still on the bank with the brakes off

3 ## Reading the formulas safely

The denominator of v_max vanishes at μ = cotθ, where no speed is safe. That is not a curiosity: it is why very steep banks are only used with high-grip tyres and why ice on a banked road is so dangerous.

4 ## A worked band

Same corner: R = 80 m, θ = 27°, μ = 0.35. Find the fastest and slowest safe speeds.

$$v_{\max}^2 = \frac{784(0.454 + 0.312)}{0.891 - 0.159} = 820 \;\Rightarrow\; 28.6\ \text{m/s}$$

$$v_{\min}^2 = \frac{784(0.454 - 0.312)}{0.891 + 0.159} = 106 \;\Rightarrow\; 10.3\ \text{m/s}$$

→ the design speed sits between them at 20 m/s, as it must

KEY RESULT

the band

$$v_{\max}^2 = \frac{Rg(\sin\theta + \mu\cos\theta)}{\cos\theta - \mu\sin\theta}$$

$$v_{\min}^2 = \frac{Rg(\sin\theta - \mu\cos\theta)}{\cos\theta + \mu\sin\theta}$$

TRAP

sign in the denominator

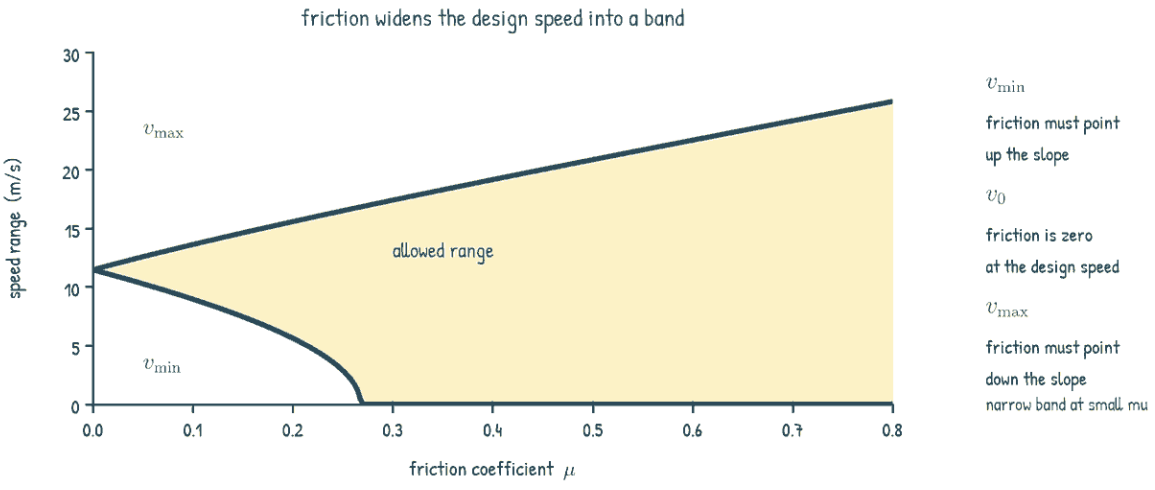
The two denominators differ: cosθ − μ sinθ for the fast limit, cosθ + μ sinθ for the slow one. Swapping them still gives plausible numbers.

CHECK

set mu = 0

Both formulas must collapse to $v_0 = \sqrt{Rg\tan\theta}$. If they do not, the algebra is wrong.

Friction does not set the design speed; it allows a range of speeds around it



$v_{\min}$

friction must point up the slope

v_0

friction is zero at the design speed

$v_{\max}$

friction must point down the slope

narrow band at small mu

KEEP / friction widens a single design speed into a band of allowed speeds

Why does a conical pendulum ignore the mass?

? A bob whirled on a string traces a horizontal circle. What actually sets the period?

1 **Separate vertical from radial**

The string makes a constant angle θ with the vertical. Vertically there is no acceleration; radially the horizontal component of the tension is the centripetal force.

$$T\cos\theta = mg, \qquad T\sin\theta = m\omega^2(L\sin\theta)$$

2 **Divide and read off the period**

The $\sin\theta$ cancels, and so does the mass. What survives is that the bob behaves like a simple pendulum of length $L\cos\theta$ — the vertical drop from the support to the plane of the circle.

$$\omega^2 = \frac{g}{L\cos\theta}, \qquad T_p = 2\pi\sqrt{\frac{L\cos\theta}{g}}$$

→ a faster whirl means a larger θ, not a different length

3 **The two limits**

As $\theta \to 0$ the circle shrinks and the period tends to the ordinary pendulum $2\pi\sqrt{(L/g)}$. As $\theta \to 90^\circ$ the required ω diverges, which is why you cannot whirl a bob into a horizontal circle.

4 **A worked whirl**

A 0.5 m string makes 40° with the vertical. Find the speed of the bob and the tension in a 0.2 kg bob.

$$T = \frac{mg}{\cos 40^\circ} = \frac{1.96}{0.766} = 2.56\ \text{N}$$

$$v = \sqrt{gL\sin\theta\tan\theta} = \sqrt{9.8(0.5)(0.643)(0.839)} = 1.63\ \text{m/s}$$

→ the period $2\pi\sqrt{(L\cos\theta/g)}$ = 1.24 s is independent of the mass

KEY RESULT

the result

$$T_p = 2\pi\sqrt{\frac{L\cos\theta}{g}}$$

$$\cos\theta = \frac{g}{\omega^2 L}$$

TRAP

using L sin theta

The radius is L $\sin\theta$, but the period uses the vertical drop L $\cos\theta$. Mixing them gives a period that grows with θ instead of shrinking.

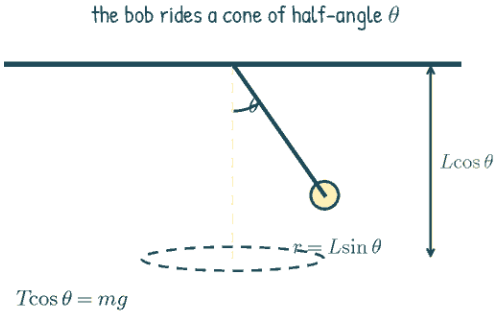
NUMBERS

a check

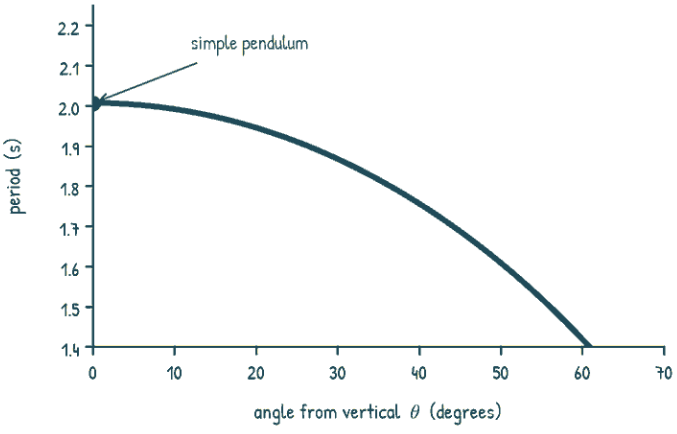
L = 1 m, θ = 60°: T = $2\pi\sqrt{(0.5/9.8)}$ = 1.42 s

the simple pendulum would give 2.01 s

The vertical component carries the weight, so the period depends on L cos(theta), never on mass



it shortens as the cone opens



KEEP / the vertical component carries the weight, so only L cos theta enters the period

What is the slowest a ball can go round a vertical loop?

? A ball on a string is whirled in a vertical circle. Why must it be fast at the bottom, and what happens at the top?

1 ## The string can pull but not push

At the top both the tension and the weight point downward, toward the centre. At the limiting speed the tension just reaches zero and the weight alone supplies mv²/R.

$$T + mg = \frac{mv^2}{R} \;\Rightarrow\; v_{\text{top, min}} = \sqrt{gR}$$

2 ## Energy carries that back to the bottom

Going from the bottom to the top costs 2mgR. Put in the minimum top speed and the required launch speed falls out; then the tension at any angle follows from the radial equation.

$$v_{\text{bot, min}} = \sqrt{5gR}$$

$$T(\theta) = 3mg(1 + \cos\theta)$$

→ θ measured from the bottom: 6mg at the bottom, 0 at the top

3 ## Losing the string

Launch slower than √(5gR) and the string slackens before the top. The ball then moves as a projectile inside the circle, and it re-tensions with a jerk if the geometry brings it back to the string.

→ the slack angle is where T = 0 and the radial equation still holds

4 ## A worked loop

A 0.4 kg ball on a 0.8 m string is whirled at exactly the limiting speed. Find the launch speed and the tension at the bottom.

$$v_{\text{bot}} = \sqrt{5gR} = \sqrt{5(9.8)(0.8)} = 6.26 \text{ m/s}$$

$$T_{\text{bot}} = 6mg = 6(3.92) = 23.5 \text{ N}$$

→ the same ball at the top has T = 0 and v = 2.8 m/s, not zero

KEY RESULT

the limiting launch

$$v_{\text{top}} = \sqrt{gR}$$

$$v_{\text{bot}} = \sqrt{5gR}$$

$$T(\theta) = 3mg(1 + \cos\theta)$$

NUMBERS

the six and the zero

T_bottom = 6mg, T_top = 0

at the horizontal position T = 3mg

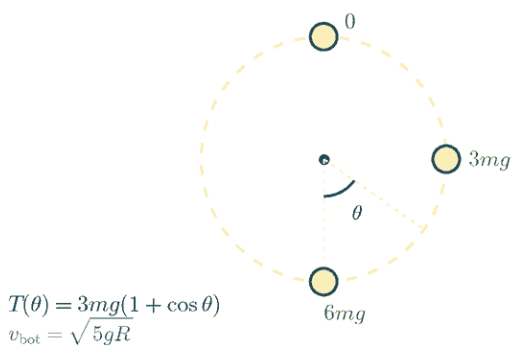
TRAP

v_top = 0

Zero speed at the top is the rod condition, not the string condition. On a string the ball must still be moving fast enough for mg to supply mv²/R.

At the limiting launch the string is just taut at the top: six mg at the bottom, zero at the top

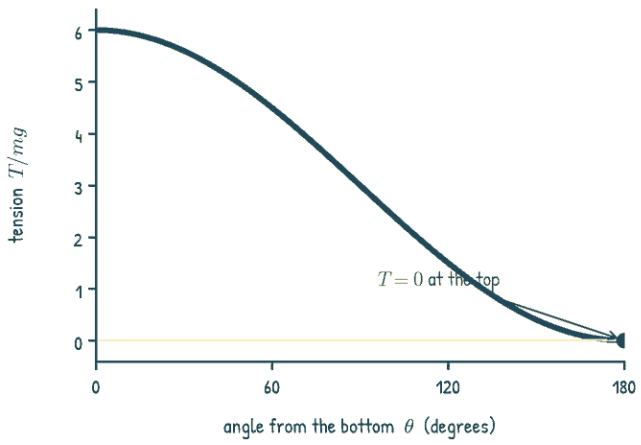
a string at the limiting speed



$T(\theta) = 3mg(1 + \cos\theta)$

$v_{\text{bot}} = \sqrt{5gR}$

tension falls all the way up



KEEP / the string can only pull, so at the top it must still be taut

Why can a bead on a rod stop at the top?

? A bead threaded on a light rod is given just enough speed to reach the highest point. What is different from the string?

1 ### A rod is a two-sided constraint

The rod can exert a force toward the centre or away from it. That means v can fall to zero at the top without the constraint failing, and the threshold √(gR) simply disappears.

$$N + mg = \frac{mv^2}{R} \quad \text{with} N \text{ of either sign}$$

2 ### Energy fixes the speed everywhere

Give the bead just enough to reach the top, v_top = 0, so v_bot = √(4gR) = 2√(gR). Then the radial equation gives N at any angle, with cosθ measured from the bottom.

$$N(\theta) = mg(2 + 3\cos\theta)$$

→ 5mg at the bottom, −mg at the top: the rod is pulling, not pushing

3 ### The sign flip

Set N = 0 and cosθ = −2/3, which is 131.8°. Below that the rod pushes outward; above it the rod pulls inward. The bead is never in danger of leaving the circle.

4 ### A worked rod loop

A 0.1 kg bead on a 0.6 m rod is given just enough speed to reach the top. Find the launch speed and N at the horizontal position.

$$v_{\text{bot}} = 2\sqrt{gR} = 2\sqrt{5.88} = 4.85 \text{ m/s}$$

$$N(90^\circ) = 2mg = 1.96 \text{ N inward}$$

→ only half the string's launch speed, because the rod does not need tension at the top

KEY RESULT

rod results

$$v_{\text{bot, min}} = 2\sqrt{gR}$$

$$N(\theta) = mg(2 + 3\cos\theta)$$

$$N = 0 \text{ at } \cos\theta = -2/3$$

TRAP

string and rod mixed

The exam favourite: check whether the constraint can push. If it can, the top speed may be zero; if it cannot, the minimum is √(gR).

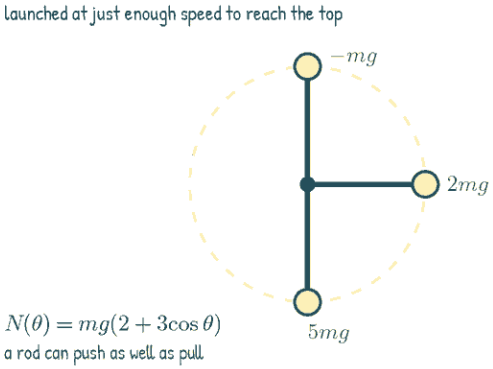
NUMBERS

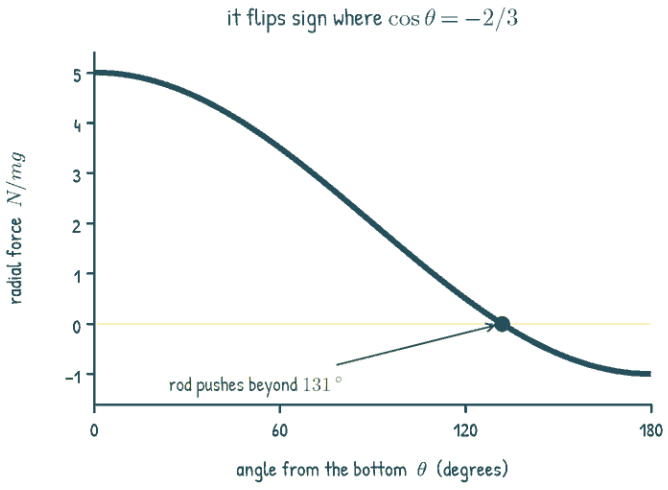
sign check

bottom: N = 5mg (outward, rod pushes)

top: N = −mg (inward, rod pulls)

A rod needs no minimum speed at the top, and past 131 degrees it stops pulling and starts pushing





KEEP / a rod pushes as well as pulls, so it needs no minimum speed at the top

Why is energy faster than forces in a vertical circle?

? The tangential equation has a sin theta in it and the radial equation has a v squared. How do you avoid integrating at all?

1 Two equations, two different jobs

The radial equation relates T and v at one instant. It says nothing about how v changes with angle. For that, use energy: weight is conservative, so the speed at any height follows from the launch speed.

$$v^2(\theta) = v_0^2 - 2gR(1+\cos\theta)$$

→ θ is measured from the bottom, so the rise is R(1 + cosθ)

2 Then one radial equation finishes it

Substitute the speed into the radial relation. The tension now comes out as a function of angle with no integration anywhere.

$$T(\theta) = m\frac{v^2(\theta)}{R} - mg\cos\theta$$

$$T(\theta) = \frac{mv_0^2}{R} - mg(2+3\cos\theta)$$

→ this single formula contains the string results 6mg and 0 at the two ends

3 Why the tangential route is a trap

You could write the tangential equation a_t = −g sinθ and integrate it with v dv = a_t R dθ. It gives the same answer, in three times the lines and with one more place to lose a sign.

→ the energy route also tells you at a glance which launches are too slow

4 A worked energy check

A 0.5 kg ball on a 1 m string is launched at 8 m/s. Use energy to find the speed at the top and the tension there.

$$v_{\text{top}}^2 = 64 - 4(9.8)(1) = 24.8 \;\Rightarrow\; v = 4.98\ \text{m/s}$$

$$T = \frac{0.5(24.8)}{1} - 0.5(9.8) = 7.5\ \text{N}$$

→ the tangential integral would give the same 24.8, in three more lines

KEY RESULT

the pair

$$v^2 = v_0^2 - 2gR(1+\cos\theta)$$

$$T = \frac{mv^2}{R} - mg\cos\theta$$

NUMBERS

a check with v0 = 8 m/s

R = 1 m, m = 0.5 kg

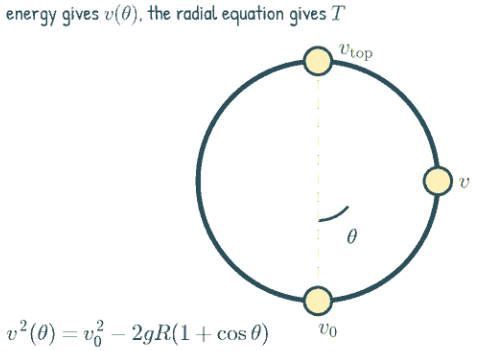
bottom T = 37 N, side 22 N, top 7 N

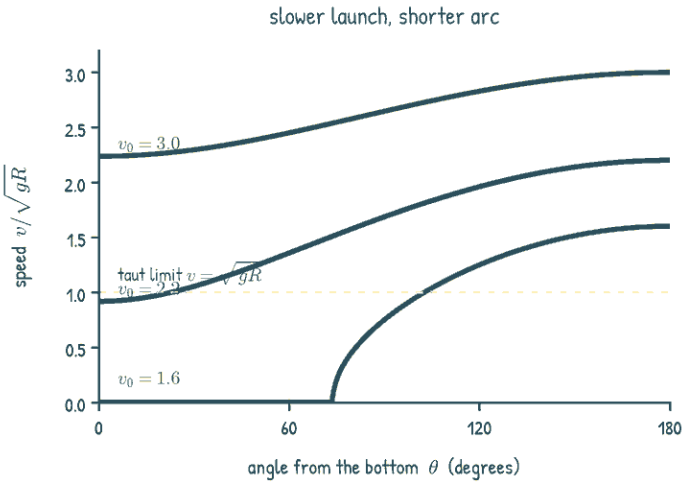
TRAP

the sign of mg cos theta

At the top the weight points inward, so the radial equation is T + mg = mv²/R. Written with cosθ from the bottom, that is exactly the minus sign above.

Energy gives the speed at every angle in one line, and the radial equation does the rest





KEEP / energy gives the speed at every angle; the radial equation gives the tension

What changes when the table is rough?

? A block on a rough table is tied over a pulley to a hanging mass. Everything is the same as the smooth case except one term. Which one?

1

Both bodies, one string

The tension is the same throughout and the magnitudes of the accelerations match. For the table block the tension pulls forward and friction holds back; for the hanging mass the weight pulls down and the tension up.

$$T - \mu_k mg = ma, \qquad Mg - T = Ma$$

2

Eliminate T

Adding the two equations removes the tension and gives the acceleration directly. Friction enters only once, as a subtraction from the drive.

$$a = \frac{(M - \mu_k m)g}{M + m}$$

→ set μ_k = 0 and the standard smooth result returns at once

3

The two thresholds

At rest the system stays put while M g ≤ μ_s m g. That gives a maximum hanging mass. Once moving, the acceleration is positive only if M > μ_k m, which is a slightly lower threshold.

$$M_{\max} = \mu_s m \quad \text{(static)}, \qquad M > \mu_k m \quad \text{(moves)}$$

4

A worked rough pulley

m = 4 kg on a table with μ_k = 0.3, M = 3 kg hanging. Find a and T.

$$a = \frac{(3 - 1.2)9.8}{7} = 2.52 \text{ m/s}^2$$

$$T = M(g - a) = 3(7.28) = 21.8 \text{ N}$$

→ T is below Mg = 29.4 N, because the hanging mass is accelerating down

KEY RESULT

the result

$$a = \frac{(M - \mu_k m)g}{M + m}$$

$$T = \frac{Mm(1 + \mu_k)g}{M + m}$$

NUMBERS

worked

m = 3 kg, M = 2 kg, μ_k = 0.4

a = (2 − 1.2)(9.8)/5 = 1.57 m/s²

T = 16.5 N, below the hanging weight

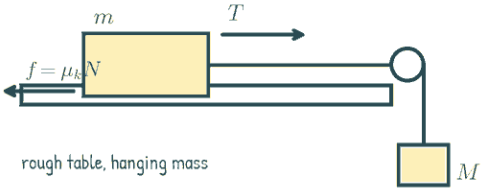
TRAP

which mu

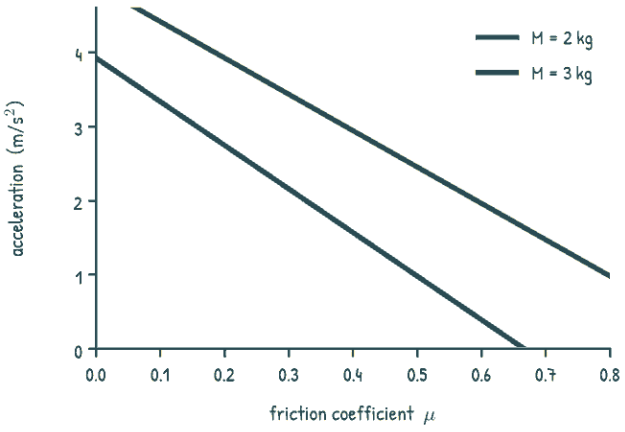
Use μ_s only for the "does it start?" question and μ_k once it moves. Mixing them gives a system that accelerates from rest with the kinetic value.

On the table friction is the brake, on the string it is the drive: separate books

one string, two different jobs



m = 3 kg on the table



KEEP / on the table friction is the brake; on the string it is the drive

Can an accelerating ramp hold a block still?

? A block sits on a frictionless ramp that is itself accelerating horizontally. Is there a ramp acceleration at which the block does not slide at all?

1 Work in the ramp frame

The block feels weight down, normal perpendicular to the face, and the pseudo force backward. Resolve along the face: the two driving terms are the downhill part of gravity and the uphill part of the pseudo force.

$$ma_{\text{rel}} = mg\sin\theta - ma_0\cos\theta$$

2 Set the relative acceleration to zero

One line gives the balanced ramp acceleration. It depends on nothing but the angle — not on m, not on g separately.

$$a_0 = g\tan\theta$$

→ θ = 30°: a_0 = 5.66 m/s² holds the block in place

3 Then the normal force

Resolve perpendicular to the face. Both the weight component and the pseudo component push the block into the ramp, so accelerating the ramp into the block increases N rather than threatening contact.

$$N = mg\cos\theta + ma_0\sin\theta > 0$$

→ this is a stability threshold, not a loss-of-contact threshold

4 A worked accelerating ramp

A 2 kg block rests on a 35° smooth ramp. Find the ramp acceleration that keeps it there and the normal force then.

$$a_0 = g\tan 35^\circ = 9.8(0.700) = 6.86\ \text{m/s}^2$$

$$N = \frac{mg}{\cos 35^\circ} = \frac{19.6}{0.819} = 23.9\ \text{N}$$

→ N is 22 % above the weight; the ramp is being pushed into the block

KEY RESULT

the balance

$$a_0 = g\tan\theta$$

$$N = m(g\cos\theta + a_0\sin\theta)$$

TRAP

confusing with liftoff

Pushing the ramp into the block raises N. Pulling the ramp away lowers it. Only the second case has a liftoff threshold, and it is a different formula.

NUMBERS

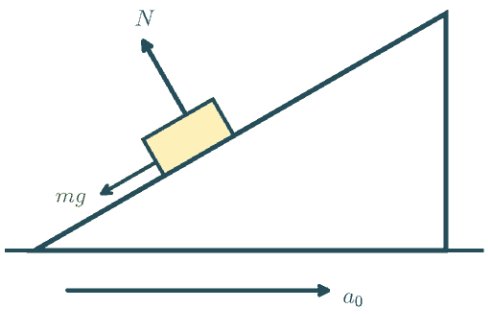
the two angles

θ = 30°: hold at 5.66 m/s²

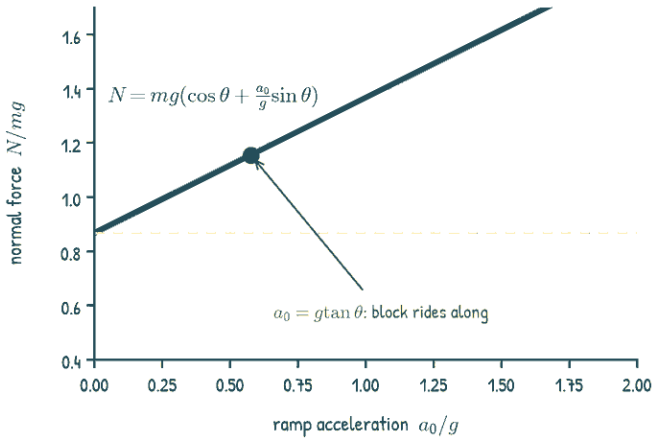
θ = 45°: hold at 9.8 m/s²

Push the ramp into the block and N rises; the block rides at rest when a0 = g tan(theta)

the ramp itself is accelerating



accelerating the ramp loads the block



KEEP / accelerate the ramp at g tan theta and the pseudo force replaces gravity along the face

How do you measure friction without a force meter?

? Friction coefficients are quoted to two digits. Which three independent experiments produce them, and what does each one assume?

1 **The tilt measures the static coefficient**

Raise one end of a plank until the block just starts to slide. At that instant the drive mg sinθ and the ceiling μ_s mg cosθ are equal, so the coefficient is the tangent of the angle.

$$\mu_s = \tan\theta_s$$

→ μ_s = 0.4 means θ_s = 21.8°, easy to read on a protractor

2 **The skid measures the kinetic one**

Lock the wheels or slide a box and measure the stopping distance from a known speed. The deceleration is μ_k g, so the coefficient follows from kinematics alone.

$$\mu_k = \frac{v^2}{2gd}$$

3 **The drag measures it a third way**

Pull at constant speed and the applied force equals the kinetic friction, so the coefficient is the ratio of two forces. This is the only one of the three that needs no kinematics, and the only one that reads the coefficient at speed.

$$\mu_k = \frac{F}{mg}$$

→ do it at several speeds: real friction is not perfectly constant

4 **A worked measurement**

A box slides 18 m before stopping from 12 m/s. Find μ_k; then find the tilt angle that would just hold it at rest if μ_s is 15 % larger.

$$\mu_k = \frac{144}{2(9.8)(18)} = 0.41$$

$$\mu_s = 0.47 \;\Rightarrow\; \theta_s = \tan^{-1} 0.47 = 25^\circ$$

→ two digits is all the data supports, and all μ is ever good for

KEY RESULT

three formulas

$$\mu_s = \tan\theta_s$$

$$\mu_k = v^2/(2gd)$$

$$\mu_k = F/(mg)$$

NUMBERS

the same surface

tilt: θ_s = 22° gives 0.40

skid from 10 m/s in 13.0 m: 0.39

drag at steady speed: 0.39

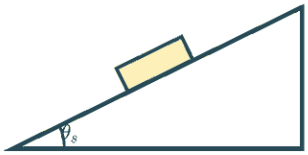
TRAP

the tilt measures mu_s only

A sliding block on the same plank is no longer at the static limit; the tangent of the angle is then just a lower bound on μ_s.

Three independent measurements of one number: tilt, brake, drag

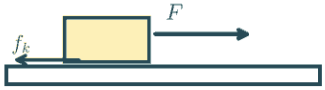
$\mu_s = \tan\theta_s$ $\mu_k = v^2/(2gd)$ $F = \mu_k mg$ at steady v



tilt until it slides



brake and measure d



drag at constant speed

KEEP / three routes to one number: tilt, brake, and drag at constant speed

Why does a chain accelerate faster as it falls?

? A chain hangs over a table edge with length l already off. The driving weight grows as more falls, but the mass being accelerated does not. What motion results?

1 **Write the force with x in it**

The hanging part has weight λxg and the whole chain has mass λL. Friction is ignored here. The acceleration therefore depends on position, which means the motion is not uniform.

$$a = \frac{\lambda x g}{\lambda L} = \frac{g x}{L}$$

2 **Integrate instead of using a constant-a formula**

Use v dv = a dx. The result is a difference of squares, not a linear ramp in time, and it shows the speed saturating only when the last link leaves.

$$v\,dv = \frac{g}{L} x\,dx \;\Rightarrow\; v^2 = \frac{g}{L}(x^2 - l^2)$$

$$v_{x=L} = \sqrt{\frac{g}{L}(L^2 - l^2)}$$

→ set l = 0 and the speed is √(gL) — the classic result

3 **The same idea with friction**

Add μ times the normal force on the part still on the table and the drive becomes x − μ(L − x). The acceleration can start negative: the chain then waits until enough hangs over.

4 **A worked chain**

A 1.5 m chain of mass 0.6 kg has 0.4 m hanging. Find the initial acceleration and the speed when it leaves the table.

$$a = \frac{gx}{L} = \frac{9.8(0.4)}{1.5} = 2.61\ \text{m/s}^2$$

$$v = \sqrt{\frac{9.8}{1.5}(2.25 - 0.16)} = 3.70\ \text{m/s}$$

→ the final speed is close to √(gL) = 3.83 because only 0.4 m was left hanging

KEY RESULT

the motion

$$a = gx/L$$

$$v^2 = \tfrac{g}{L}(x^2 - l^2)$$

NUMBERS

a check

L = 2 m, l = 0.5 m

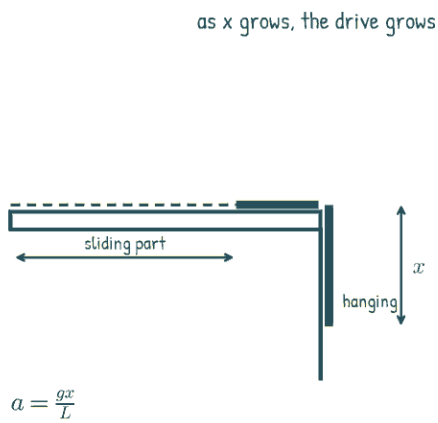
v at the end = √(9.8(3.75)/2) = 4.3 m/s

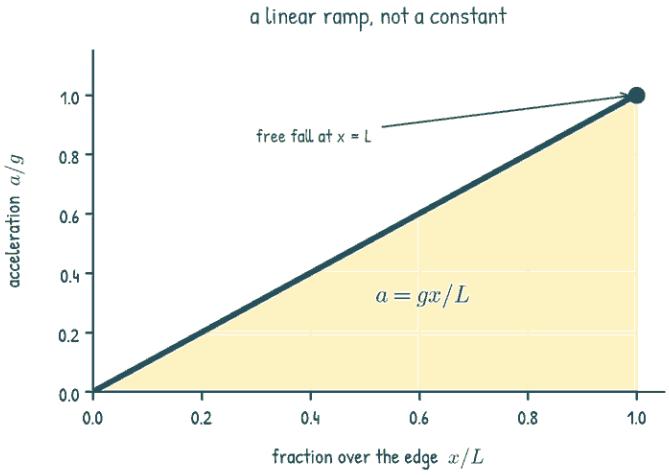
TRAP

constant acceleration

Using a = gx/L as a constant with s = ½at² gives the wrong speed and the wrong time. The acceleration is a function of position.

The hanging weight grows while the sliding mass stays the same, so the acceleration grows with x





KEEP / the hanging weight grows while the sliding mass stays the same, so a grows with x

How hard must you push a wedge to carry its block?

? A block rests on a smooth wedge. Accelerate the wedge horizontally and the block can stay put. What is the required acceleration, and what keeps it there?

1 The two ways to hold the block

On a fixed smooth wedge the block slides down, because nothing balances mg sinθ. Accelerating the wedge adds a pseudo force whose uphill component can balance it exactly.

$$ma_0\cos\theta = mg\sin\theta \;\Rightarrow\; a_0 = g\tan\theta$$

2 What the rider feels

The normal force rises, because the pseudo force also presses the block into the face. That is not a coincidence: the same push that replaces gravity along the slope also loads the contact.

$$N = \frac{mg}{\cos\theta}$$

→ a 1 kg block on a 30° wedge at the balance point presses with 11.3 N, not 8.5 N

3 The applied force on the whole system

Treat wedge plus block as one body. The external push must accelerate the total mass at a_0, and the internal normal does not appear at all.

$$F = (M+m)g\tan\theta$$

4 A worked carry

A 2 kg block on a 40° wedge is to ride along. Find the required acceleration and the push on a 5 kg wedge.

$$a_0 = 9.8\tan 40^\circ = 8.22\ \text{m/s}^2$$

$$F = (5+2)(8.22) = 57.6\ \text{N}, \qquad N = \frac{19.6}{\cos 40^\circ} = 25.6\ \text{N}$$

→ the push scales with the total mass; the balance angle does not

KEY RESULT

the balance

$$a_0 = g\tan\theta$$

$$N = mg/\cos\theta$$

$$F = (M+m)g\tan\theta$$

NUMBERS

worked

M = 4 kg, m = 1 kg, θ = 30°

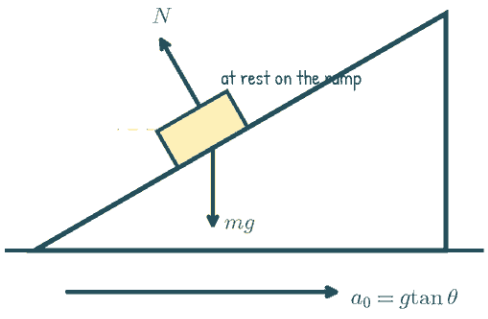
a_0 = 5.66 m/s², F = 28.3 N

TRAP

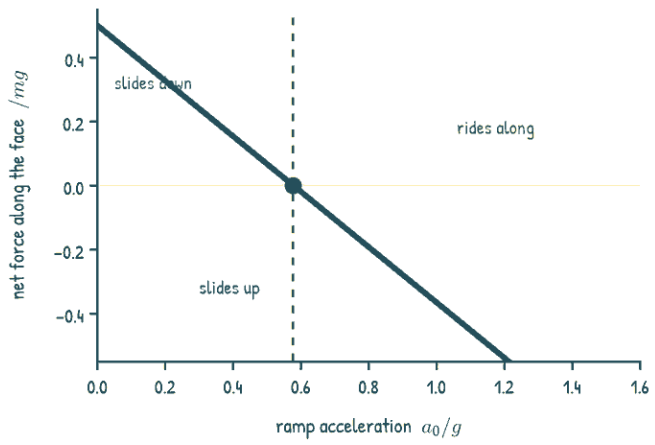
pushing the wedge the other way

Reverse the acceleration and the pseudo force now adds to the downhill pull: no speed can hold the block. Direction is part of the answer.

At a0 = g tan(theta) the pseudo force exactly replaces the downhill pull of gravity



the residual slides the block one way or the other



KEEP / at a0 = g tan theta the pseudo force exactly replaces the downhill pull

Is there friction on a block tied to the axis?

? A block on a turntable is tied by a string to the centre and the table spins fast. Is the friction zero, or maximum?

1 **First ask what the motion needs**

The block travels a circle of radius r at angular speed ω, so it needs an inward force mω²r. That requirement is fixed by the kinematics and has nothing to do with which agent supplies it.

$$F_{\text{in}} = m\omega^2 r$$

2 **Two suppliers, one budget**

Without the string the only inward force is friction, so f = mω²r and the block slips when that exceeds μ_s mg. With a taut string, the string can carry the whole requirement and friction falls to zero.

$$T + f = m\omega^2 r$$

→ a string that is taut and radial does not need help from friction

3 **Both frames must agree**

In the rotating frame the block is at rest with a centrifugal term mω²r outward. The string pulls inward, friction adjusts to whatever is left, and equilibrium gives the same answer. The frames agree because they must.

4 **A worked turntable**

m = 0.5 kg at r = 0.4 m on a table spinning at 6 rad/s, string taut to the axis, μ_s = 0.3. Is there friction, and which way?

$$\text{needed}: \ m\omega^2 r = 0.5(36)(0.4) = 7.2 \text{ N}$$

$$\text{ceiling}: \ \mu_s mg = 1.47 \text{ N}$$

→ the string alone cannot supply 7.2 N, so friction must add 1.47 N inward — and it slips

DEFINE

the budget rule

Compute the required inward force first. Then share it between the agents present, in the order they can act.

TRAP

"friction always opposes motion"

Here friction would have to point inward, with the motion, to keep the block in its circle. It is a centripetal supplier, not a drag.

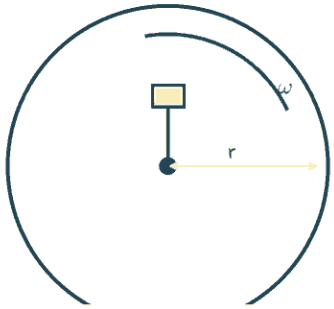
INSIGHT

why the puzzle exists

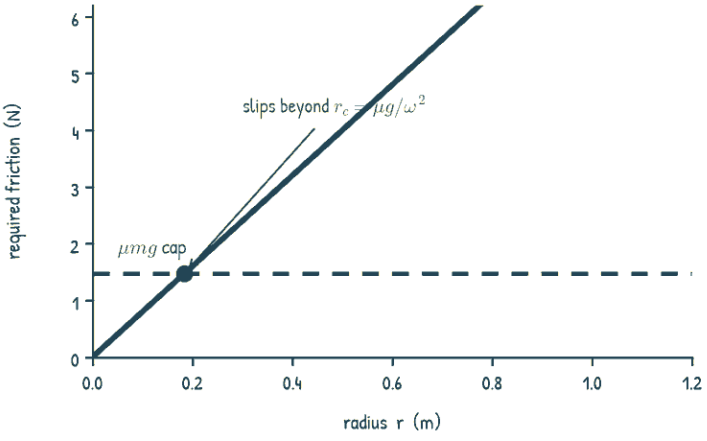
Students assume friction must be doing something because the surface is rough and spinning. It is doing nothing: the string has already met the requirement.

A block on a turntable stays only while friction can supply m omega squared r

top view: block tied to the axis



friction supplies the centripetal force



KEEP / the string and friction share one job, so each takes only what it must

Why does a coin and a brick slip at the same speed?

? Two objects of very different mass sit at the same radius on a spinning turntable. Which one slides off first?

1 ## Set the requirement equal to the ceiling

Contact survives while friction can supply the centripetal requirement. Both sides carry a factor m, so the mass drops out of the inequality entirely.

$$m\omega^2 r \le \mu_s m g \;\Rightarrow\; \omega \le \sqrt{\frac{\mu_s g}{r}}$$

2 ## So the outer edge goes first

At fixed ω the limit is a radius: everything inside r_c stays, everything outside leaves. That is why a record on a turntable throws the dust at its rim, and why a coin placed further out survives a shorter spin-up.

$$r_c = \frac{\mu_s g}{\omega^2}$$

3 ## What breaks the tie

Mass cancels only because both the requirement and the ceiling are proportional to it. Tilt the table, or add a string, or make one object tall enough to topple, and the cancellation is gone.

→ a tall block topples before it slides: that is a torque condition, not a force one

4 ## A worked spin-up

A coin sits at r = 12 cm on a turntable with μ_s = 0.4. At what angular speed does it slide, and does a 2 kg brick at the same radius differ?

$$\omega_{\max} = \sqrt{\frac{0.4(9.8)}{0.12}} = 5.72 \text{ rad/s}$$

$$m \text{ cancels: the brick slides at the same } \omega$$

→ the coin reaches the rim first only because it is placed further out, not because it is lighter

KEY RESULT

the limit

$$\omega_{\max} = \sqrt{\mu_s g / r}$$

$$r_c = \mu_s g / \omega^2$$

NUMBERS

a check

μ_s = 0.3, r = 0.5 m

ω_max = 2.42 rad/s

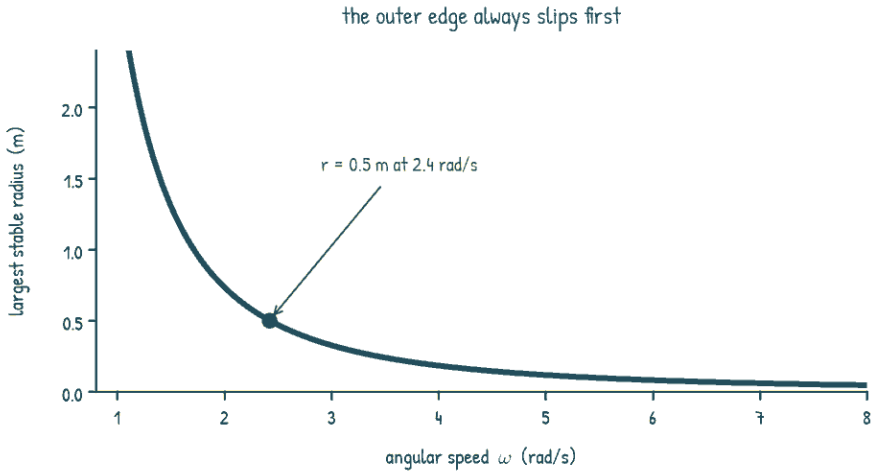
mass never enters

TRAP

heavier sticks better

It does not. The heavier object needs more inward force, and gets exactly proportionally more friction ceiling.

Without a string, friction alone must supply m omega squared r, so the limit is independent of mass



$$m\omega^2 r \le \mu_s m g$$

$$\omega_{\max} = \sqrt{\mu_s g / r}$$

mass cancels

so a coin and a brick slip together

if placed at the same r

KEEP / the friction limit is proportional to mass, so mass cancels out of the slip condition

Why is the block's acceleration not along the face?

? A wedge slides on a table and a block slides down its face. You know both motions are linked, but how?

1 **Write the constraint in coordinates**

Fix a lab frame. The block's horizontal position is the wedge position plus its offset along the face; its height depends only on how far it has slid. One parameter, s, describes the block.

$$x_b = x_w - s\cos\theta, \qquad y_b = y_0 - s\sin\theta$$

2 **Differentiate twice**

The constraint is linear in the coordinates, so differentiating twice gives the same relation between accelerations. There is nothing to guess: the accelerations are locked by geometry.

$$a_{bx} = a_w - a_s\cos\theta, \qquad a_{by} = -a_s\sin\theta$$

3 **Then let the forces do their job**

Only when the kinematics is written down do the equations close: the normal force is the one unknown interaction, and the wedge's own acceleration is driven by its horizontal component. For a smooth wedge and block released from rest,

$$a_w = -\frac{mg\sin\theta\cos\theta}{M + m\sin^2\theta}$$

→ the wedge runs opposite to the block's horizontal slide, and the sign is worth checking

4 **A worked wedge**

M = 4 kg wedge at 30°, m = 1 kg block, all smooth. Find the wedge acceleration.

$$a_w = -\frac{mg\sin\theta\cos\theta}{M + m\sin^2\theta} = -\frac{9.8(0.433)}{4.25}$$

$$a_w = -1.00\ \text{m/s}^2, \qquad a_{\text{rel}} = \frac{5(9.8)(0.5)}{4.25} = 5.76\ \text{m/s}^2$$

→ the wedge recoils at 1.0 m/s² while the block slides down the face at 5.8

GATE

the order of operations

1 coordinates 2 constraint 3 differentiate 4 forces 5 solve

Writing forces first is what produces the wrong signs.

KEY RESULT

results for a smooth wedge

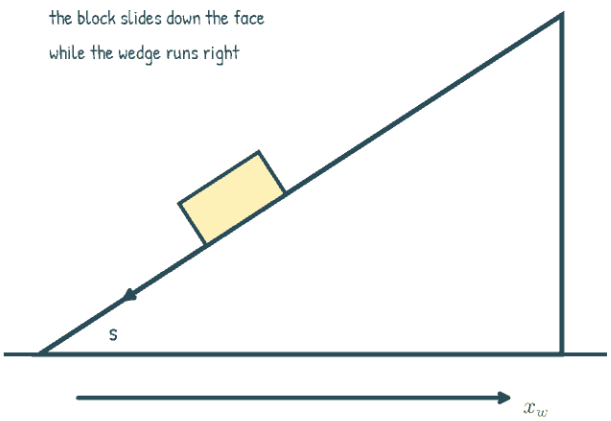
$$a_{\text{rel}} = \frac{(M+m)g\sin\theta}{M + m\sin^2\theta}$$

TRAP

assuming a_block along the face

In the lab frame the block does not move along the face: the face itself is moving. Only the relative acceleration is along the face.

A contact is a geometric constraint: it links coordinates before it links forces



$x_b = x_w - s\cos\theta$

$y_b = y_0 - s\sin\theta$

one constraint links the two coordinates

differentiate twice for a

$a_b \neq a_w$

KEEP / a contact is a coordinate constraint first and a force second

What is g inside a lift, a car, a centrifuge?

? A pendulum in an accelerating vehicle keeps time differently. How much, and in which direction does it hang?

1 One vector replaces gravity

In a frame accelerating at a_0 every body feels a pseudo acceleration $-a_0$. Combine it with g and the pair behaves exactly like a new, stronger, tilted gravity.

$$\vec{g}_{\text{eff}} = \vec{g} - \vec{a}_0, \qquad |\vec{g}_{\text{eff}}| = \sqrt{g^2 + a_0^2}$$

2 Everything follows from the new g

A plumb line hangs along g_eff, a water surface is perpendicular to it, and a pendulum swings with the period set by its magnitude. Nothing else changes.

$$T = 2\pi\sqrt{\frac{L}{g_{\text{eff}}}} = T_0\left(1 + \frac{a_0^2}{g^2}\right)^{-1/4}$$

→ at $a_0 = g$ the period is 0.84 T_0 — the same factor as √2 to the power −1/2

3 The lift cases

Upward acceleration adds to g, downward subtracts, free fall zeroes it. In free fall a pendulum does not swing at all, because there is no "down" left inside the cabin.

4 A worked lift

A 1 m pendulum rides a lift accelerating upward at 3 m/s². Find g_eff and the period; repeat for free fall.

$$g_{\text{eff}} = 12.8\ \text{m/s}^2, \qquad T = 2\pi\sqrt{1/12.8} = 1.76\ \text{s}$$

$$\text{free fall}: \quad g_{\text{eff}} = 0, \qquad T \to \infty$$

→ a pendulum in free fall does not swing at all: there is no preferred down

KEY RESULT

vector rule

$$\vec{g}_{\text{eff}} = \vec{g} - \vec{a}_0$$

$$T = T_0\left(1 + a_0^2/g^2\right)^{-1/4}$$

NUMBERS

the 0.84

$a_0 = g$: g_eff = 1.414 g

T = 0.84 T_0

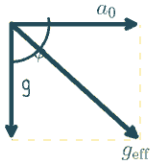
TRAP

scalar addition

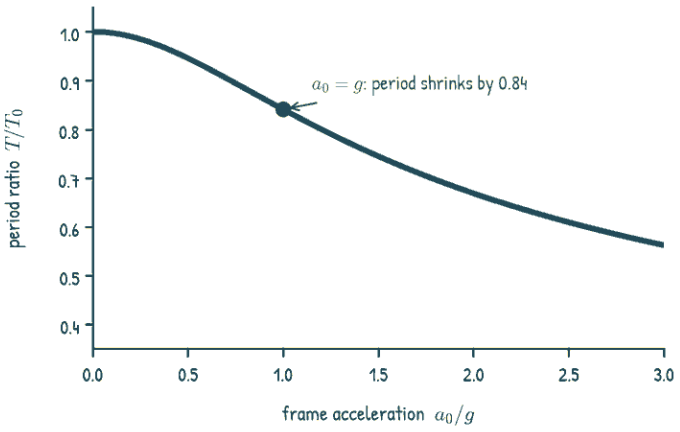
g and a_0 add as vectors. Only in the lift (vertically aligned) may you add or subtract their magnitudes.

Inside an accelerating frame, gravity is replaced by a longer vector tilted by arctan(a0/g)

gravity and the frame add as vectors



the pendulum speeds up as the car pulls away



KEEP / g_eff is the vector sum of gravity and the frame acceleration, taken with a minus sign

When does a block leave a wedge that is pulled away?

? The wedge is yanked horizontally out from under a block resting on its face. At what acceleration does contact break?

1 Use the wedge frame

In the wedge frame the block feels its weight, the pseudo force ma_0 opposite the pull, and the normal. Resolve perpendicular to the face: that direction has no acceleration while contact lasts.

$$N = mg\cos\theta - ma_0\sin\theta$$

2 Contact dies at N = 0

A normal force cannot be negative. Setting N to zero gives the threshold acceleration, and beyond it the block leaves the surface and becomes a projectile in the lab frame.

$$a_0 = g\cot\theta$$

→ θ = 30°: threshold 1.73g

3 The first instant after liftoff

At the threshold the block's acceleration perpendicular to the face is exactly the free-fall component, so it just separates. The lab-frame description switches from constrained motion to projectile motion at that instant.

4 A worked liftoff

A 45° wedge is pulled horizontally until the block leaves. Find the threshold and the normal force at $a_0 = 0.5g$.

$$a_0 = g\cot 45^\circ = g = 9.8\ \mathrm{m/s^2}$$

$$N = mg(\cos 45^\circ - 0.5\sin 45^\circ) = 0.354mg$$

→ at half the threshold the block still presses with about a third of its weight

KEY RESULT

threshold

$$N = mg\cos\theta - ma_0\sin\theta$$

$$a_0 = g\cot\theta \text{ forliftoff}$$

TRAP

pulling toward the face

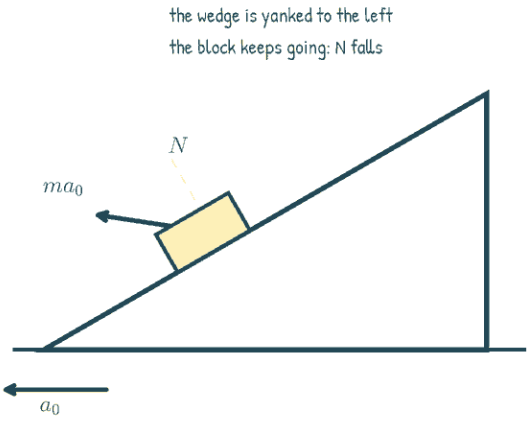
Accelerate the wedge into the block and N rises. Only pulling the surface away from the block can make N fall.

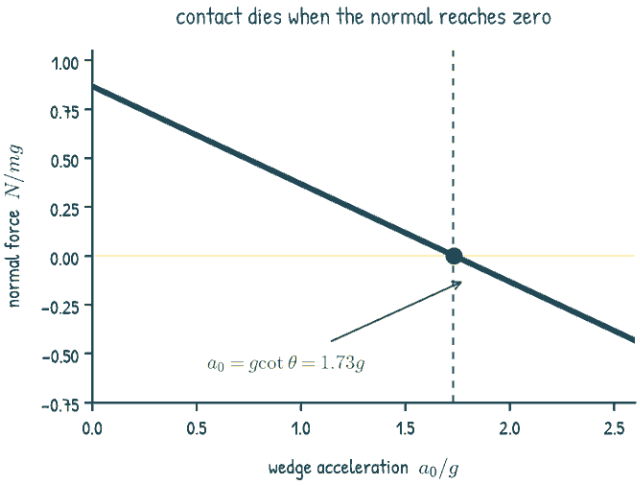
CHECK

θ near 90 degrees

A vertical face needs an enormous pull to separate the block: $a_0 \to 0$ as $\theta \to 90°$, which is the sense of $\cot\theta$.

If the wedge is pulled out from under the block, the normal force falls to zero at a0 = g cot(theta)





KEEP / contact survives only while the normal force remains positive

Why can you ignore the normal force in a work calculation?

? A constraint force can be large, yet contributes nothing to the energy bookkeeping. Why exactly?

1 **Displacement allowed by the constraint**

Give the system a small imaginary displacement that respects the constraints. The normal force is perpendicular to the surface, and the sliding is along it, so the dot product vanishes.

$$\delta W = N \cdot \delta s = 0$$

2 **The wedge, solved without forces**

The block stays on the face, so its vertical displacement is zero to first order while it slides. Setting the total virtual work of gravity to zero gives the relation between the wedge displacement and the slide directly.

$$\delta y_b = \delta x_w \cos\theta - \delta s \sin\theta = 0$$

→ one equation, and no free-body diagram at all

3 **Where the method pays**

Linked systems with several constraints: the internal forces never appear, so the number of equations collapses. The price is that you must write the kinematics correctly — the method is intolerant of a wrong sign.

4 **A worked constraint**

The block on a 30° wedge is held fixed while the wedge is pushed 2 cm. How far does the block rise, and what force does the wedge feel?

$$\delta y_b = \delta x_w \tan\theta = 2(0.577) = 1.15 \text{ cm}$$

$$N = \frac{mg\cos\theta}{1 + \frac{m}{M}\sin^2\theta}$$

→ one constraint, one line, and no free-body diagram was drawn

GATE

when to use it

Equilibrium or energy questions in a linked system, where you want a relation between displacements rather than internal forces.

TRAP

friction is not a constraint force

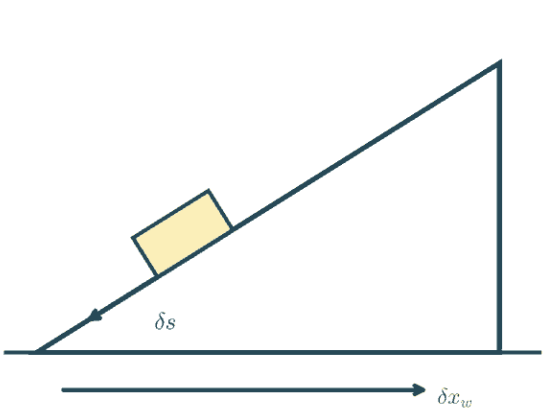
Friction along the surface does work, and the work is negative. Virtual work only deletes the ideal constraints: normals and taut ropes.

INSIGHT

the deep reason

Constraint forces are defined to be perpendicular to the allowed motion. That single geometric fact is why energy methods are shorter than force methods.

Internal contact forces do no work along the constraint, which is why virtual work can skip them



$\delta W = N \cdot \delta s = 0$

$N \perp \delta s$

a normal force does no work when the sliding is along the surface

so the constraint never appears in the work

KEEP / a constraint force perpendicular to the allowed displacement does no work

At what length does a rope over an edge start to slide?

? A rope lies on a rough table with length x hanging over the edge. How much must hang before it moves?

1 **Two weights, one friction**

Model the rope as linear density λ. The hanging part pulls with λxg; the part on the table is held by friction up to μλ(L−x)g. Motion begins when the pull beats the ceiling.

$$\lambda x g = \mu \lambda (L - x) g$$

2 **Solve for the fraction**

The density cancels, which is why the answer is a pure geometry statement: a rope starts to slide when the hanging fraction reaches μ/(1+μ).

$$\frac{x}{L} = \frac{\mu}{1+\mu}$$

→ μ = 0.4: a little over 28 % must hang

3 **Once it moves**

Friction is now kinetic and the driving weight grows as more rope goes over, so the acceleration increases with x. The motion is not uniform, and energy is the easier route to the speed.

$$a = \frac{g[x - \mu_k(L-x)]}{L}$$

4 **A worked rope**

A 2 m rope with μ_s = 0.3 hangs over a table edge. How much must hang before it moves, and what is the initial acceleration?

$$\frac{x}{L} = \frac{0.3}{1.3} = 0.231 \;\Rightarrow\; x = 46 \text{ cm}$$

$$a = g[x - \mu_k(1-x)] = 9.8(0.231 - 0.231) = 0$$

→ at the threshold the acceleration is zero by construction, and then grows with x

KEY RESULT

threshold

$$x/L = \mu_s/(1+\mu_s)$$

NUMBERS

values

μ_s = 0.2: 16.7 %

μ_s = 0.4: 28.6 %

smooth table: any length moves it

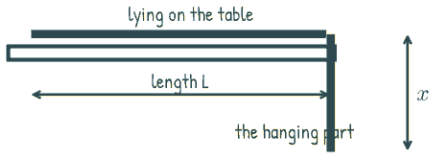
TRAP

using the whole weight

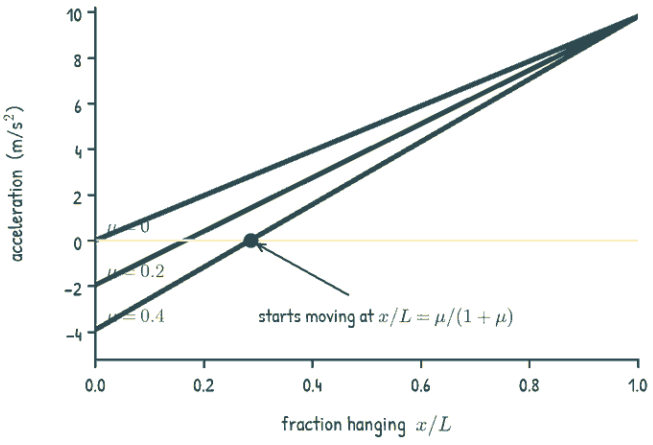
Friction acts only on the part still on the table. Using the full weight λLg overstates the ceiling and predicts no motion.

A rope over an edge accelerates only once the hanging fraction beats friction on the rest

the hanging part pulls, the rest drags



friction holds it until enough hangs



KEEP / the hanging fraction must beat friction on the part still on the table

Does a falling chain weigh three times as much?

? A chain is dropped so that it lands in a heap on a scale. At the instant the last link lands, what does the scale read?

1 **Two jobs, two terms**

The scale supports the weight of what has already landed and also destroys the momentum of what is arriving. Both are forces on the scale, so they add.

$$F = \lambda x g + \dot{m}\, v = \lambda x g + \lambda v^2$$

2 **Use free fall to eliminate v**

The moving part has fallen a distance x, so $v^2 = 2gx$. Substitute and the momentum term becomes twice the weight of the pile, whatever x is.

$$F = \lambda x g + 2\lambda x g = 3\lambda x g$$

→ at the instant the last link lands, x = L and F = 3mg

3 **Why the printed answer says 2mg**

The source computes only the momentum flux, or only the weight, at the final instant and reports 2mg. The correct bookkeeping has both terms present simultaneously, so the reading is 3mg. This is the one place where the edition deliberately departs from the source; see SOURCE-AUDIT.

4 **A worked falling chain**

A 2 kg chain of length 1 m lands in a heap. What does the scale read at the instant the last link lands?

$$F = 3\lambda L g = 3mg = 3(19.6) = 58.8 \text{ N}$$

$$\text{source answer}: \; 2mg = 39.2 \text{ N}$$

→ the weight term and the momentum term are both present; the source dropped one

KEY RESULT

the two terms

$$F = \lambda g x + \lambda v^2$$

$$v^2 = 2gx \;\Rightarrow\; F = 3\lambda g x$$

TRAP

dropping a term at the end

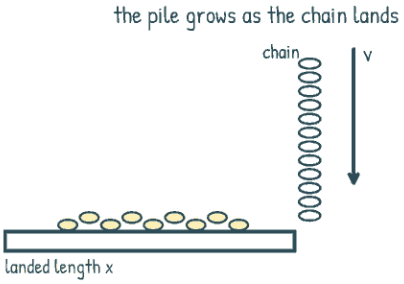
At the final instant neither term vanishes. The pile still has weight and the last link still has momentum to lose.

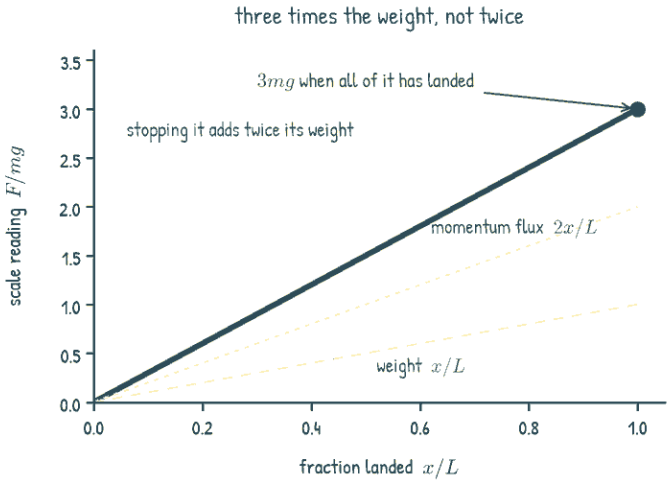
INSIGHT

the general pattern

Any process that catches mass gives weight plus momentum flux. The same two terms give 3mg here and λv² in the lifting problem.

A landing chain must be stopped as well as held: the scale reads 3x the landed weight





KEEP / a landing chain must be stopped as well as held: three times the landed weight

Why is the cost of speed logarithmic?

? A rocket reaches 9 km/s with an exhaust speed of 3 km/s. Why must it be mostly fuel, and how much?

1 ## Momentum balance in the rocket frame

In the instantaneous rest frame the rocket throws mass backward at v_e. Momentum conservation over one small burn gives the differential relation.

$$m\,dv = -v_e\,dm \;\Rightarrow\; dv = -v_e\frac{dm}{m}$$

2 ## Integrate

The right side integrates to a logarithm, which is the whole reason rockets are staged. Equal steps of speed cost equal ratios of mass, not equal masses.

$$\Delta v = v_e \ln\frac{m_0}{m_f}$$

$$\frac{m_0}{m_f} = e^{\Delta v/v_e}$$

3 ## Put in the numbers

Nine kilometres per second with a three kilometre per second exhaust needs a mass ratio of e³ ≈ 20. Nineteen twentieths of the launch mass is propellant, before any structure or payload.

$$\frac{m_0}{m_f} = e^{3} \approx 20$$

→ burning half the mass gives only 0.693 v_e, not half of the maximum

4 ## A worked burn

A rocket with v_e = 2.5 km/s must reach 7 km/s. What mass ratio is needed, and how much of it can be payload?

$$\frac{m_0}{m_f} = e^{7/2.5} = e^{2.8} = 16.4$$

$$\text{payload fraction} \le \frac{1}{16.4} = 6.1\ \%$$

→ two stages at the same total fuel reach further, because the empty tank is dropped

KEY RESULT

the equation

$$\Delta v = v_e \ln(m_0/m_f)$$

NUMBERS

ratios worth knowing

half the mass burned: 0.693 v_e

90 % burned: 2.30 v_e

99 % burned: 4.61 v_e

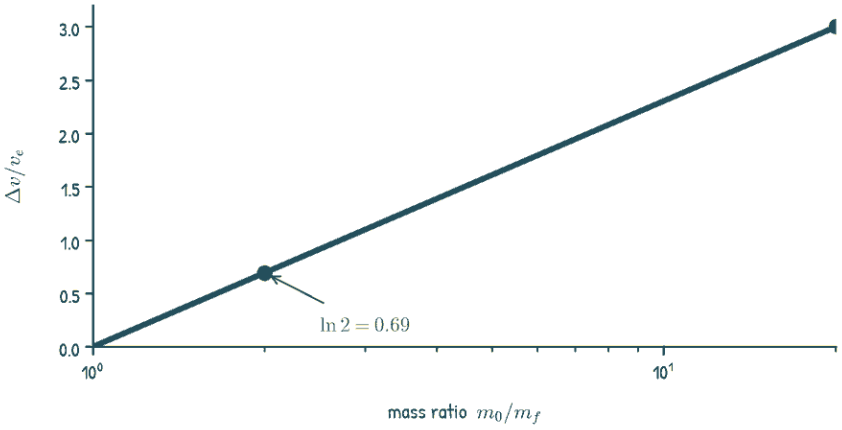
TRAP

averaging speeds

You cannot average the exhaust speed over the burn; the rocket is lighter later, so late mass counts more. That is exactly what the logarithm encodes.

Every extra km/s costs exponentially in mass: the rocket equation is a budget, not a force

the cost of speed is logarithmic



$$m\,dv = -v_e\,dm$$

$$\Delta v = v_e \ln\frac{m_0}{m_f}$$

9 km/s at 3 km/s exhaust:

$$m_0/m_f = e^{3} \approx 20$$

95 % of the launch mass is propellant

KEEP / each unit of speed costs a fixed ratio of mass, so speed grows as a logarithm

Why does a conveyor belt need twice the power?

? Sand falls vertically onto a belt moving at constant speed. The belt's motor supplies power. Where does it go?

1 **Force from momentum flux**

The sand arrives with no horizontal momentum and leaves with v. The belt must supply momentum at rate $\dot{m}v$, so the force it exerts is $\dot{m}v$.

$$F = \frac{dp}{dt} = \dot{m}v$$

2 **Power in, energy out**

The belt moves at v, so the motor supplies P = Fv = $\dot{m}v^2$. But the sand only gains kinetic energy at rate $\frac{1}{2}\dot{m}v^2$ — the energy equation has a gap.

$$P_{\text{motor}} = \dot{m}v^2, \qquad \dot{K} = \tfrac{1}{2}\dot{m}v^2$$

→ the missing half is dissipated as the sand slips on the belt before it grips

3 **Where the heat comes from**

The sand slides relative to the belt until it is up to speed. Kinetic friction at the interface does work on the sand, and the sliding itself generates heat. The bookkeeping closes exactly at half.

$$Q = \tfrac{1}{2}\dot{m}v^2$$

4 **A worked conveyor**

Sand falls at 12 kg/s onto a belt moving at 1.5 m/s. Find the driving force, the motor power, and the heat generated.

$$F = \dot{m}v = 18\ \text{N}, \qquad P = Fv = 27\ \text{W}$$

$$\dot{K} = \tfrac{1}{2}\dot{m}v^2 = 13.5\ \text{W}, \qquad Q = 13.5\ \text{W}$$

→ half the motor's output leaves as heat, every second, forever

KEY RESULT

three rates

$$F = \dot{m}v$$

$$P = \dot{m}v^2$$

$$\dot{K} = \tfrac{1}{2}\dot{m}v^2$$

TRAP

efficiency of 50 % is not a fault

No belt can do better while the sand slips. To approach 100 % you must accelerate the sand gently on a slower belt first.

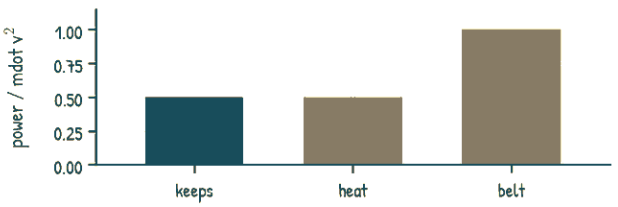
INSIGHT

same algebra elsewhere

Rope picked up from the floor, chain landing on a scale, water hitting a wall: all have a momentum-flux force and a factor-of-two energy story.

Accelerating arriving mass costs twice the kinetic energy: the other half is friction heat





$$F = \dot{m}v$$

$$P = Fv = \dot{m}v^2$$

$$\dot{K} = \tfrac{1}{2}\dot{m}v^2$$

the missing half is heat

KEEP / half the power becomes kinetic energy, half is lost to friction while the sand slips

Why does lifting a chain cost more than its weight?

? A chain is pulled up from a heap at constant speed. The force is larger than the weight of the hanging part. By how much?

1 ### Two terms again

The lifted length x has weight λxg. New links are being started from rest at the rate λv, and each must be given speed v, so an extra force λv² is needed.

$$F = \lambda x g + \lambda v^2$$

2 ### Power and the excess

The power delivered is Fv. Of that, λxgv raises potential energy and λv³ goes into kinetic energy of freshly moving links. Since those links end up with ½λv³ each second, the other half of the flux term is lost to the inelastic jerk.

$$P = Fv = \lambda x g v + \lambda v^3, \qquad \dot{K}_{\text{new}} = \tfrac{1}{2}\lambda v^3$$

3 ### Accelerating as well

If v is not constant there is a third term from the changing speed of the already lifted part. Constant speed isolates the momentum flux, which is why textbooks specify it.

4 ### A worked lift

A 0.8 kg/m chain is lifted at 1.2 m/s. Find the force when 2.5 m hangs, and the rate of energy loss.

$$F = \lambda g x + \lambda v^2 = 19.6 + 1.15 = 20.8 \text{ N}$$

$$P_{\text{lost}} = \tfrac{1}{2}\lambda v^3 = 0.69 \text{ W}$$

→ the extra force is only 6 % of the weight at this speed, but it never goes away

KEY RESULT

the force

$$F = \lambda g x + \lambda v^2$$

NUMBERS

a check

λ = 0.4 kg/m, v = 1.5 m/s, x = 3 m

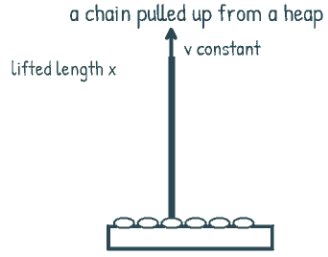
weight 11.8 N, flux 0.9 N, total 12.7 N

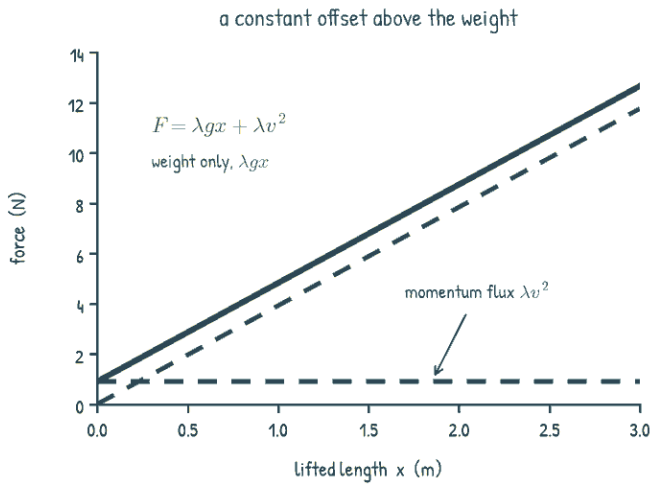
TRAP

constant speed means constant force

The force is constant in v but grows with x. A graph of F against x is a straight line offset above the weight line.

Lifting at constant speed still costs an extra momentum-flux force, lambda v squared





KEEP / lifting at constant speed still costs a momentum-flux force, lambda v squared

Why can one hand hold a ship?

? A rope takes two turns round a bollard and a small force holds a huge one. What is the law, and where does the ratio come from?

1 A small element of rope

Take a piece subtending $d\theta$. The tension changes by dT across it, and the normal force on the element is $T\,d\theta$. Friction can supply at most μ times that normal, so the tension can only change gradually.

$$dT = \mu T\, d\theta$$

2 Integrate round the wrap

The differential equation separates, and the solution is exponential in the wrap angle. That is why the effect explodes with turns: 2π radians is already a factor of e^{2πμ}.

$$\frac{T_2}{T_1} \le e^{\mu\theta}, \qquad \theta = 2\pi N$$

→ μ = 0.3, one turn: a factor of 6.6

3 Why diameter does not matter

The radius cancels out of the derivation: a thin carabiner and a thick bollard hold equally well for the same μ and the same angle. Only the total wrap and the surface matter.

4 A worked bollard

A rope takes two turns round a bollard with μ = 0.25. What load can a 200 N pull hold?

$$T_2 = T_1 e^{\mu\theta} = 200 e^{0.25(4\pi)} = 200(23.1)$$

$$T_2 = 4.6 \text{ kN}$$

→ a third turn takes it to 22 kN and a fourth to 107 kN: the growth is exponential

KEY RESULT

the capstan law

$$T_2 = T_1 e^{\mu\theta}$$

$$\theta = 2\pi N$$

NUMBERS

turns

one turn (μ = 0.3): 6.6 : 1

two turns: 43 : 1

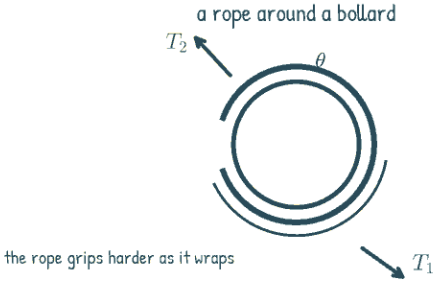
three turns: 287 : 1

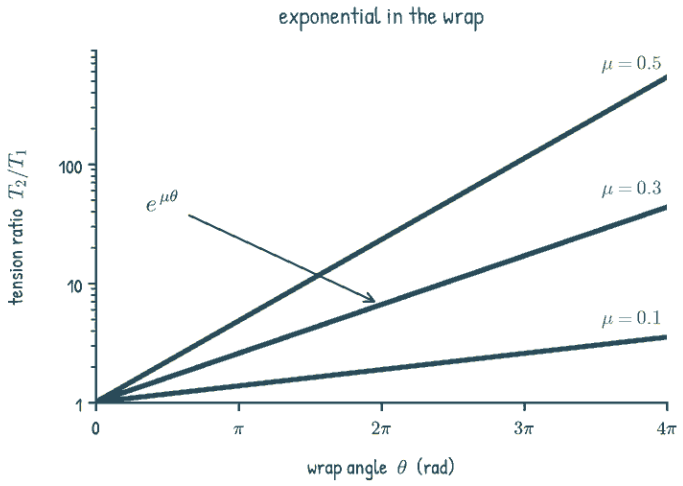
TRAP

the rope must not slip

The exponential is an upper bound, attained only at the point of slipping. Below it the ratio can be anything smaller.

Every radian of wrap multiplies the grip by e to the mu theta





KEEP / every radian of wrap multiplies the grip by e to the mu theta

How does a friction hitch hold a falling climber?

? A rope wrapped round a carabiner holds with a small hand force. How many turns does it take to turn 1 into 10, and into 300?

1

Same law, inverted

The capstan relation applies unchanged: the hand holds the low-tension end and the load is the high-tension end. Solve for the wrap needed at a given ratio.

$$\theta = \frac{1}{\mu}\ln\frac{T_2}{T_1}, \qquad N = \frac{\theta}{2\pi}$$

2

Numbers for a real rope

With μ = 0.3, a factor of ten needs 1.22 turns; a factor of 287 needs three. The curve is logarithmic, so the first turn buys most of the available holding power.

$$N = \frac{\ln 10}{2\pi(0.3)} = 1.22 \text{ turns}$$

→ three turns at μ = 0.3: e^{5.66} = 287

3

Why hitches fail in the wet

Water or ice drops μ, and the ratio is exponential in μ, so a small loss of grip is a large loss of holding. That is why the same number of turns is safe on dry nylon and marginal on a wet rope.

4

A worked hitch

A wet rope has μ = 0.18 and must hold 8 kN with a 400 N hand. How many turns are needed?

$$N = \frac{\ln(20)}{2\pi(0.18)} = \frac{3.00}{1.13} = 2.65 \text{ turns}$$

$$\text{dry, } \mu = 0.3: \quad N = 1.59 \text{ turns}$$

→ wetness costs a whole extra turn, which is why hitches are checked in the rain

KEY RESULT

turns for a ratio

$$N = \frac{\ln(T_2/T_1)}{2\pi\mu}$$

NUMBERS

μ = 0.3

10 : 1 at 1.22 turns

100 : 1 at 2.44 turns

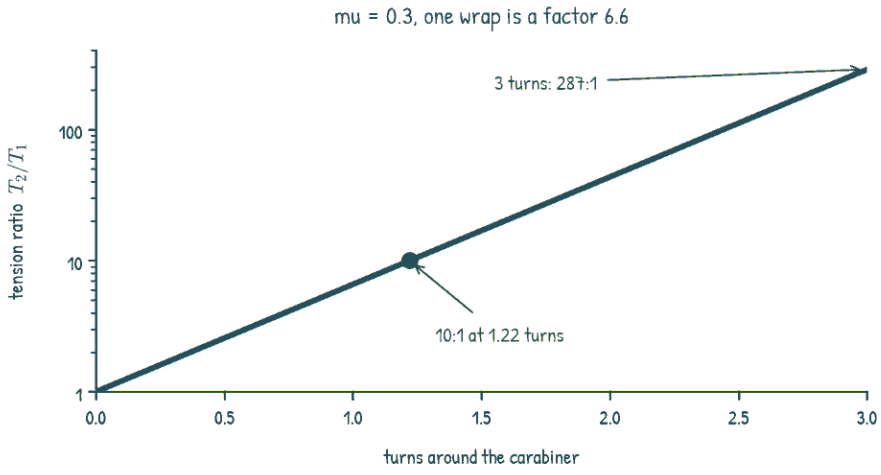
287 : 1 at 3 turns

INSIGHT

why it feels like magic

The hand only sets the low-tension end. Friction along the wrap does the holding, and it grows exponentially rather than linearly.

A friction hitch holds because tension decays exponentially along the wrap, not because you grip harder



$T_2 = T_1 e^{\mu\theta}$

$\theta = 2\pi N$

friction does the holding, the hand only sets the load

works for any rope diameter

KEEP / the hitch holds because tension decays exponentially along the wrap

Why does staging beat one big tank?

? Two rockets carry the same fuel and the same structure, but one drops an empty stage. How much speed does that buy?

1 The rocket equation is multiplicative

Speed adds across stages while mass ratios multiply. Dropping dry structure raises the mass ratio of the second burn, and the logarithm turns that into extra speed.

$$\Delta v = v_e \ln \frac{m_0}{m_1} + v_e \ln \frac{m_1'}{m_f}$$

2 A worked comparison

Launch 1000 kg with 800 kg of fuel, 100 kg payload and two 50 kg stages, exhaust 2 km/s. One stage must carry both dry masses to the end; two stages drop one of them halfway.

$$\mathrm{two\,stages}: 2\ln\tfrac{1000}{600} + 2\ln\tfrac{550}{150} = 1.02 + 2.60 = 3.62$$

$$\mathrm{one\,stage}: 2\ln\tfrac{1000}{200} = 3.22\ \mathrm{km/s}$$

→ 0.40 km/s for free, from the same fuel and the same hardware

3 Why not infinitely many stages

Each stage adds its own dry mass and its own engine. Beyond a few stages the structure you add costs more than the dead mass you drop, so real vehicles settle at two or three.

4 A worked two-stage

Same rocket, but the second stage is only 40 kg dry. How much does the extra 10 kg saved buy?

$$\Delta v_2 = 2\ln\tfrac{540}{140} = 2.70\ \mathrm{km/s}$$

$$\Delta v = 1.02 + 2.70 = 3.72\ \mathrm{km/s}\ (\mathrm{was}\ 3.62)$$

→ 10 kg of structure is worth 0.10 km/s: the logarithm makes dry mass expensive

KEY RESULT

the sum

$$\Delta v = v_e \sum_i \ln \frac{m_{0,i}}{m_{f,i}}$$

NUMBERS

the worked case

stage 1: 1000 → 600 kg, 1.02 km/s

stage 2: 550 → 150 kg, 2.60 km/s

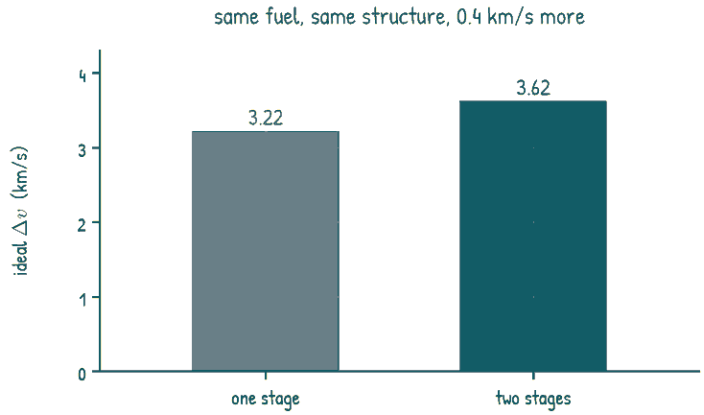
single stage: 3.22 km/s

TRAP

adding speeds without dropping mass

If the empty stage is carried to the end, staging buys nothing at all. The gain is entirely from the drop.

Staging wins by throwing away dead structure before the next burn



stage 1

1000 → 600 kg

Δv = 1.02 km/s

drop 50 kg

550 → 150 kg

Δv = 2.60 km/s

single stage

1000 → 200 kg

KEEP / staging wins by throwing away dead structure before the next burn

When is it safe to ignore gravity during a hit?

? A bat hits a ball for about a millisecond. Gravity never switches off — so why does every solution drop it?

1

Compare impulses, not forces

Gravity acts the whole time, but its impulse over the contact window is tiny compared with the bat's. What matters for the velocity change is the integral, so the ratio of impulses is the test.

$$\frac{J_g}{J_{\text{bat}}} = \frac{mg\,\Delta t}{J_{\text{bat}}}$$

2

Put in the numbers

A 0.145 kg ball, 1 ms, and a 1 N·s bat impulse: gravity contributes 1.4 mN·s, which is 0.14 % of the total. The error from dropping it is smaller than the timing uncertainty.

$$J_g = (0.145)(9.8)(10^{-3}) = 1.4 \times 10^{-3}\ \text{N s}$$

→ 0.14 % — the impulse approximation is quantitative, not a hand wave

3

When it fails

A long contact, a light force, or a slow process: a balloon resting on your hand, a block sliding down a ramp. Then gravity's impulse is comparable and must be kept.

4

A worked window

A 0.06 kg tennis ball is in contact with the strings for 4 ms under a 300 N average force. How much does gravity change the answer?

$$J_{\text{hit}} = 1.2\ \text{N s}, \qquad J_g = 2.4 \times 10^{-3}\ \text{N s}$$

$$\frac{J_g}{J_{\text{hit}}} = 0.2\ \%$$

→ a 4 ms contact is still short enough; a 0.2 s contact would not be

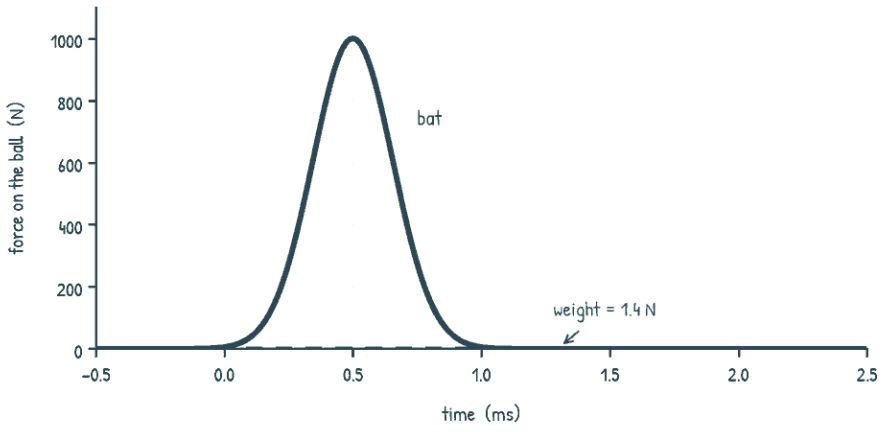
GATE
the test
Compute mg Δt and compare it with the impulse of the external hit. Below about 1 % it is safe to drop.

NUMBERS
the window
m = 0.145 kg, Δt = 1 ms
J_g = 1.4 × 10⁻³ N·s
J_bat = 1 N·s → 0.14 %

TRAP
dropping gravity from the energy
Ignoring gravity during the hit is legitimate. Ignoring the height change of the ball afterwards is not: the same problem may need both.

A force that is huge for a millisecond makes gravity irrelevant; that is the whole impulse approximation

same window, two forces, 700 times apart



$J_{\text{bat}} = 1\ \text{N s}$

$J_g = 1.4 \times 10^{-3}\ \text{N s}$

ratio 0.14 %

so gravity is dropped for the duration of the hit

KEEP / a force that is huge for a millisecond makes gravity irrelevant

Why does a spinning hoop push its bead up the wire?

? A bead slides on a hoop spinning about its vertical diameter. Below a critical spin it sits at the bottom. What happens above it?

1 Balance along the wire

In the rotating frame the bead feels weight, the normal from the wire, and the centrifugal term. The wire can only push perpendicular to itself, so the tangential components must cancel.

$$\sin\theta(g - \omega^2 R\cos\theta) = 0$$

2 Read off the two families of solutions

The first factor gives the bottom, always an equilibrium. The second gives a pair of symmetric positions that exist only when the spin is fast enough for the cosine to be less than one.

$$\cos\theta = \frac{g}{\omega^2 R}, \qquad \omega_c = \sqrt{g/R}$$

→ at ω = 2ω_c the bead sits at 75.5° from the bottom

3 Stability, not just equilibrium

Below ω_c the bottom is stable and the off-axis solutions do not exist. Above it the bottom becomes unstable — a pitchfork bifurcation — and the bead climbs to one of the two new seats.

$$\omega < \sqrt{g/R} \;\Rightarrow\; \theta = 0 \text{ stable}$$

4 A worked hoop

R = 0.5 m hoop spinning at 6 rad/s. Where does the bead settle, and what happens at 3 rad/s?

$$\omega_c = \sqrt{g/R} = 4.43 \text{ rad/s}$$

$$\cos\theta = \frac{9.8}{36(0.5)} = 0.544 \;\Rightarrow\; \theta = 57^\circ$$

→ at 3 rad/s, below ω_c, the only stable seat is the bottom

KEY RESULT

the equilibria

$$\cos\theta = g/(\omega^2 R)$$

$$\omega_c = \sqrt{g/R}$$

NUMBERS

a check

R = 1 m, ω_c = 3.13 rad/s

ω = 2ω_c: cosθ = 0.25, θ = 75.5°

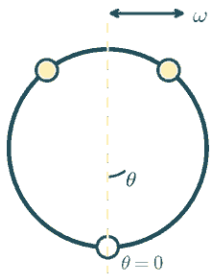
TRAP

calling the bottom unstable below the threshold

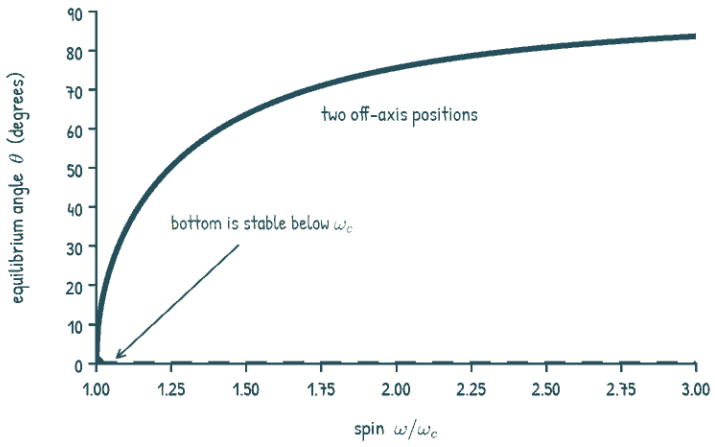
It is stable below and unstable above. The word "always" is what makes this question a discriminator.

Below the critical spin the bead sits at the bottom; above it the bottom becomes unstable and two new seats appear

the hoop spins about its vertical diameter



a pitchfork in the equilibrium



KEEP / above the critical spin the bottom becomes unstable and two new seats appear

At what angle does a slackening string let go?

? Launch a ball round a vertical circle too slowly and the string goes limp before the top. Is the release angle predictable?

1 ## Two conditions at the same instant

At the moment the string slackens the tension is zero and the ball is still on the circle. Both must be true together: T = 0 gives one equation, energy gives the speed at that angle.

$$\frac{mv^2}{R} = mg\cos\theta_0, \qquad v^2 = v_0^2 - 2gR(1+\cos\theta_0)$$

2 ## Eliminate the speed

Substituting removes v and leaves a single equation for the angle. The result is worth remembering: it depends only on the dimensionless launch speed $v_0^2/(gR)$.

$$\cos\theta_0 = \frac{v_0^2}{3gR} - \frac{2}{3}$$

→ $v_0 = \sqrt{5gR}$ gives $\cos\theta_0 = 1$: the release is exactly at the top

3 ## What happens next

Past θ_0 the string is slack and the ball is a projectile. It usually re-tensions with a jerk somewhere on the far side, unless the geometry carries it clear. Energy alone cannot tell you which.

→ the highest point of the projectile arc is not the same as the top of the circle

4 ## A worked slack

A 0.2 kg ball on a 0.9 m string is launched at 4.5 m/s. At what angle does the string slacken, and what is the speed there?

$$\cos\theta_0 = \frac{20.25}{3(9.8)(0.9)} - \frac{2}{3} = 0.765 - 0.667 = 0.098$$

$$\theta_0 = 84.4^\circ, \qquad v = \sqrt{gR\cos\theta_0} = 0.93 \text{ m/s}$$

→ the release is just above the horizontal, far below the top

KEY RESULT

slack angle

$$\cos\theta_0 = \frac{v_0^2}{3gR} - \frac{2}{3}$$

no real solution means the string never slackens

NUMBERS

a check

v_0 = 3 m/s, R = 1 m

$\cos\theta_0$ = 0.306 − 0.667 = −0.361, θ_0 = 111°

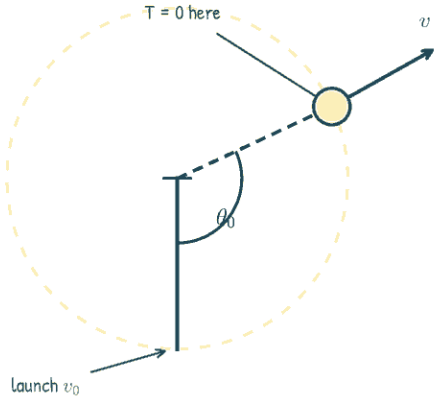
TRAP

using the top as the release point

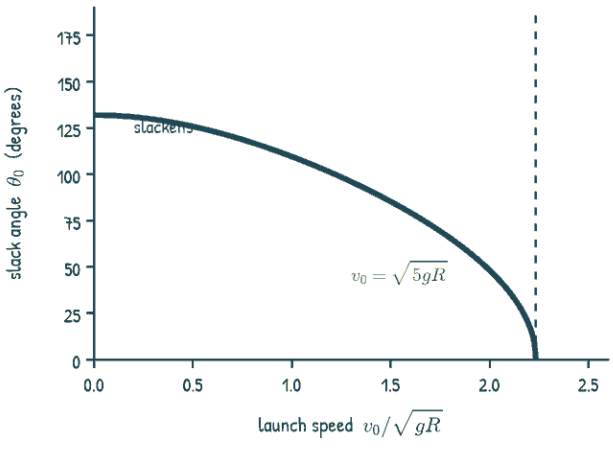
The string does not wait until the top. It goes slack wherever T first reaches zero, which for a slow launch is well before the highest point.

The slack angle is where T reaches zero, and it approaches 131 degrees as the launch gets slow

the string cannot push, so it lets go



lower launch, earlier slack



KEEP / the slack angle is where the tension reaches zero while the radial equation still holds

How far does a bouncing ball travel before it stops?

? A ball is dropped from height h and returns to e squared times that height after each bounce. The bounces are infinite — is the total distance?

1 Write the series

The first fall is h. Every later bounce contributes a rise and a fall, each shorter by a factor e^2. The sum is geometric, so a closed form exists.

$$s = h + 2h(e^2 + e^4 + \dots) = h\frac{1+e^2}{1-e^2}$$

2 The time has its own series

Each flight time scales by e, not e^2, because time goes as the square root of height. That gives a different ratio and a different closed form.

$$t = t_0\frac{1+e}{1-e}, \qquad t_0 = \sqrt{\frac{2h}{g}}$$

→ e = 0.8, h = 5 m: 46 m and 9 s

3 The two limits

As e → 1 both sums diverge: a perfectly elastic ball bounces forever. As e → 0 the distance tends to h and the time to a single fall, which is the inelastic case. Both limits are physically right, which is a good sign for the formulas.

$$e \to 1:\ s \to \infty,\ t \to \infty \qquad e \to 0:\ s \to h,\ t \to t_0$$

4 A worked bounce

A ball dropped from 2 m rebounds to 1.28 m. Find e, the total distance and the total time to rest.

$$e = \sqrt{1.28/2} = 0.8, \qquad s = 2\tfrac{1.64}{0.36} = 9.1\ \text{m}$$

$$t = \sqrt{0.408}\cdot\tfrac{1.8}{0.2} = 0.639(9) = 5.8\ \text{s}$$

→ the first fall alone takes 0.64 s; the infinite tail adds five more

KEY RESULT

the two sums

$$s = h\frac{1+e^2}{1-e^2}$$

$$t = \sqrt{\frac{2h}{g}}\cdot\frac{1+e}{1-e}$$

NUMBERS

worked

e = 0.8, h = 5 m, g = 10

s = 5(1.64/0.36) = 22.8 m

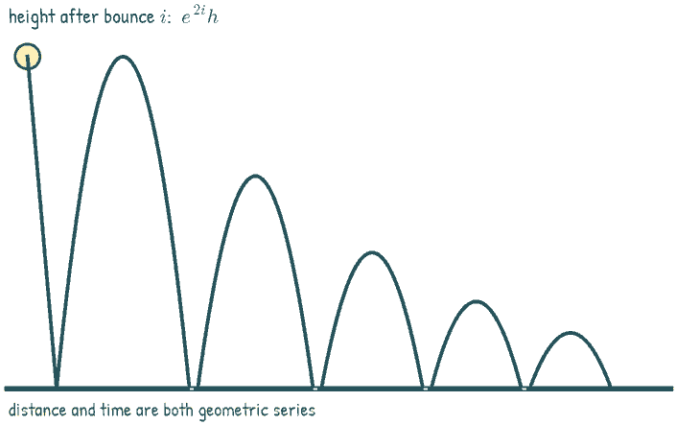
t = 1(1.8/0.2) = 9.0 s

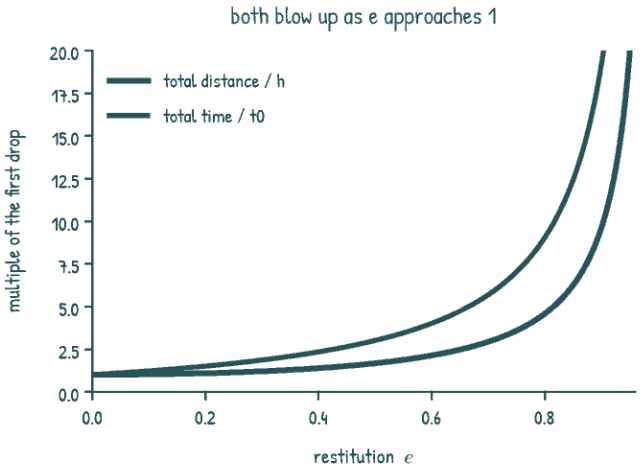
TRAP

summing heights only

Every bounce after the first contributes twice its height, once up and once down. Forgetting the factor of two halves the answer.

A bouncing ball is an infinite series with a finite sum: both distance and time close up in e





KEEP / an infinite number of bounces still sums to a finite distance and a finite time

Is horizontal momentum really conserved on a wedge?

? A block slides down a smooth wedge that is free to move on a smooth table. Which quantity is conserved, and how do you check it in one line?

1

The system is isolated horizontally

Weight and the table normal are vertical, so nothing external acts horizontally on wedge plus block. The horizontal momentum of the pair is therefore constant, and it starts at zero.

$$m(v_{bx}) + Mv_w = 0$$

2

The constraint supplies the velocities

The block stays on the face, so its horizontal velocity is the wedge velocity plus the horizontal part of its slide. Substituting into the momentum statement gives the wedge recoil directly.

$$v_{bx} = v_w - v_{\text{rel}}\cos\theta$$

$$v_w = \frac{mv_{\text{rel}}\cos\theta}{M+m}$$

→ the wedge moves opposite to the block horizontal slide, always

3

Differentiate for the accelerations

Since the relation holds at every instant, differentiating gives the same relation between accelerations. The result matches the force derivation, which is the check the exam wants.

$$a_w = -\frac{mg\sin\theta\cos\theta}{M+m\sin^2\theta}$$

4

A worked momentum check

m = 2 kg block, M = 3 kg wedge, θ = 40°, smooth. Find the wedge acceleration and check momentum in one line.

$$a_w = -\frac{2(9.8)(0.643)(0.766)}{3+2(0.413)} = -\frac{9.65}{3.83} = -2.52\ \text{m/s}^2$$

$$a_{\text{rel}} = \frac{5(9.8)(0.643)}{3.83} = 8.22\ \text{m/s}^2$$

→ check: m(a_rel cosθ + a_w) = 2(6.30 − 2.52) = 7.56 = −M a_w, so the pair stays put

KEY RESULT

the conservation statement

$$m(a_{\text{rel}}\cos\theta - a_w) = Ma_w$$

one line, and it fixes the sign of the recoil

NUMBERS

a check

m = 1 kg, M = 4 kg, θ = 30°, g = 9.8

a_w = −1.00 m/s², a_rel = 5.76 m/s²

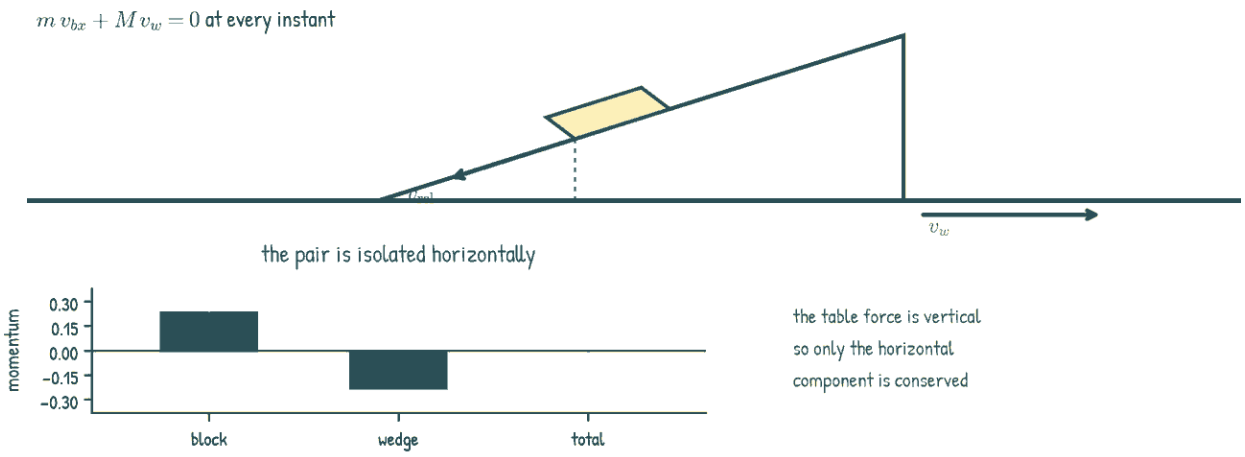
m a_bx = 1(4.99 − 1.00) = 3.99, M a_w = −4.00

TRAP

vertical momentum is not conserved

The table pushes up, so vertical momentum is not conserved. Only the horizontal component is, and only while the table stays level.

No external horizontal force means the recoil is exactly the block horizontal momentum, reversed



KEEP / no external horizontal force means the recoil is the block horizontal momentum, reversed

Where should the fuel go between two stages?

? Two stages, 800 kg of fuel, 50 kg of dry structure each. Put 400 kg in each, or 160 in the first and 640 in the second?

1

Two burns, two mass ratios

Each burn is its own Tsiolkovsky step. The dry stage dropped between them carries no Δv of its own; it only lowers the mass that the second burn must push.

$$\Delta v = v_e \ln \frac{1000}{1000 - 800f} + v_e \ln \frac{950 - 800f}{150}$$

→ f is the fraction of the 800 kg that burns in stage one

2

The curve has no interior peak

Differentiate and the result is monotone: every kilogram moved from stage one to stage two raises the total. Physically, the 50 kg dropped early is worth more the sooner it goes, so an empty first stage is best in this model.

$$f = 0.5: \ 3.62 \text{ km/s}, \qquad f = 0.2: \ 3.67, \qquad f = 0.8: \ 3.50$$

3

So why does anyone split the fuel?

Because the first stage must lift the rest off the pad, so it needs thrust and therefore propellant, and because gravity losses punish a long slow burn. The ideal Δv model has no such constraint, and the source says exactly this: the optimum depends on the dry-mass model you impose.

What survives is the size of the effect. Staging at all is worth about 0.40 km/s here; moving fuel between the stages within the sensible range is worth about 0.17.

4

A worked split

Same rocket: 1000 kg, 100 kg payload, two 50 kg dry stages, 800 kg of fuel, v_e = 2 km/s. Put 160 kg in stage one instead of 400.

$$\Delta v_1 = 2 \ln \frac{1000}{840} = 0.35, \qquad \Delta v_2 = 2 \ln \frac{790}{150} = 3.32$$

$$\text{total } 3.67 \text{ km/s versus } 3.62 \text{ at the equal split}$$

→ a small gain, and the single-stage rocket with the same structure gets 3.22

KEY RESULT

add the logs

$$\Delta v = v_e \sum_i \ln \frac{m_{0,i}}{m_{f,i}}$$

each term is one burn; the drop sits between the terms

NUMBERS

the numbers

equal split 3.62 km/s

little fuel first 3.69, most fuel first 3.22

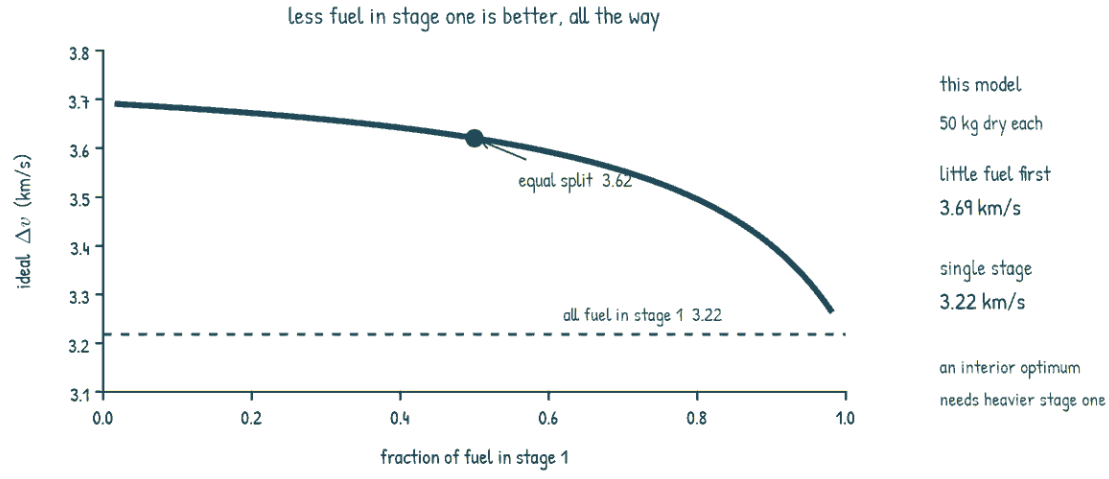
single stage with the same structure 3.22

INSIGHT

what staging buys

Not a better engine and not a better fuel. It buys the right to stop carrying an empty tank: 0.40 km/s here.

In this ideal model less fuel in stage one is always better: the gain is the dropped dry mass



KEEP / with fixed dry masses the split has no interior optimum: fuel early is never worth it

Where exactly do Newton's laws stop being true?

? Newtonian mechanics is taught as if it were exact. Under what conditions does it fail, and what replaces it?

1 ### Three fences, three replacements

Speeds near c need special relativity; actions near ħ need quantum mechanics; strong gravitational fields need general relativity. Each fence is a statement about a dimensionless ratio becoming of order one.

$$v/c \ll 1, \qquad S/\hbar \gg 1, \qquad \Phi/c^2 \ll 1$$

2 ### What survives inside the fence

Every result in this edition assumes rigid bodies, inextensible massless strings, and dry friction obeying Amontons' laws. Real strings stretch, real bodies deform, and real friction depends weakly on speed.

None of those corrections matter for a block on a ramp. All of them matter for a rope in a climbing fall, which is why the capstan numbers are quoted as upper bounds.

3 ### A model you can state the limits of

The honest statement is not "Newton was wrong". It is "Newton is exact in this regime, and here is the boundary". A solution that names its own assumptions can be checked; one that does not cannot.

4 ### A worked ratio

A proton at 0.1c, a dust grain of 10^{-15} kg moving at 1 mm/s, and a clock on a neutron star. Which need which theory?

$$v/c = 0.1 \ (\gamma = 1.005), \qquad S/\hbar \approx 10^{19}$$

$$\Phi/c^2 \approx 0.2 \text{ at the surface}$$

→ a 0.5 % relativistic correction, no quantum correction, and general relativity non-negotiable

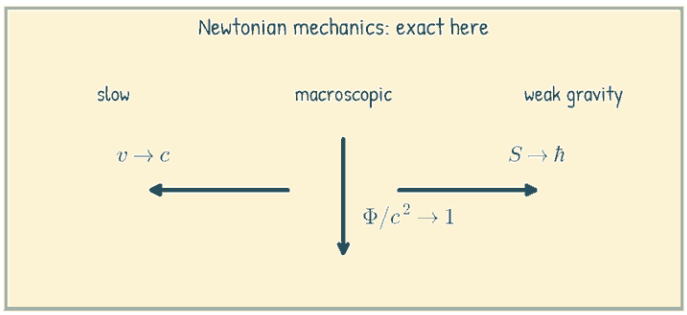
GATE
the three ratios
v/c, S/ħ, and Φ/c². Compute whichever is available and check it is small.

TRAP
quoting friction as exact
μ is empirical. Quoting a friction answer to four figures is a claim about the surface that the surface cannot support.

INSIGHT
why this is on a formula sheet
The validity column of the source table is the part examiners reward and students skip. Two marks often live there.

Newton is exact inside a fence, and knowing where the fence is is what makes the model honest

the fence is a specification, not a failure



what replaces it

$v \to c$

special relativity

clocks, lengths, $E = mc^2$

$S \to \hbar$

quantum mechanics

action in units of ħ

$\Phi/c^2 \to 1$

general relativity

gravity is geometry

KEEP / Newton is exact inside a fence, and knowing the fence is what makes the model honest

If a rolling wheel is not sliding, what is friction doing?

? A driven wheel has friction at the contact, yet the contact point does not move. How can a force act at a point at rest?

1 ## Pure rolling is a kinematic condition

The contact point has zero velocity because the centre moves forward at ωR while the rim moves backward at ωR. No slip, so the friction is static, and static friction may take any value up to its ceiling.

$$v_{cm} = \omega R, \qquad v_{\text{contact}} = 0$$

2 ## So friction can point either way

On a driven wheel the axle pushes forward and friction at the road points forward too, providing the reaction that accelerates the car. On a braked wheel friction points backward. On a freely rolling wheel it is nearly zero.

$$f \le \mu_s N$$

→ the engine does not push the car: the road does, through static friction

3 ## Where rolling resistance comes from

Rolling resistance is not the same phenomenon. It comes from deformation of the tyre and the road, which moves the normal force forward of the axle and creates a retarding torque. That is why it scales with load and with tyre pressure, not with μ.

4 ## A worked wheel

A 1200 kg car accelerates at 3 m/s² on μ_s = 0.7 tarmac. What friction does each driven wheel need, and is it available?

$$f_{\text{total}} = Ma = 3600 \text{ N}$$

$$\text{available}: \ \mu_s Mg = 0.7(11760) = 8232 \text{ N}$$

→ 44 % of the available grip, and all of it is static: no sliding anywhere

DEFINE

pure rolling

No relative motion at the contact, so v_cm = ωR and the friction is static.

TRAP

mu_k for a rolling wheel

Using kinetic friction on a wheel that is rolling correctly is a category error. Kinetic friction appears only once the wheel is skidding.

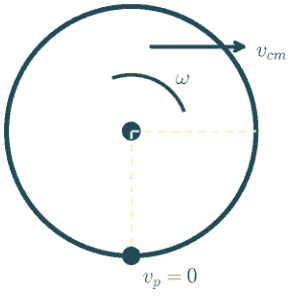
INSIGHT

the strong version

A car accelerates because the road pushes it. The engine only spins the wheels; the interaction with the ground does the rest.

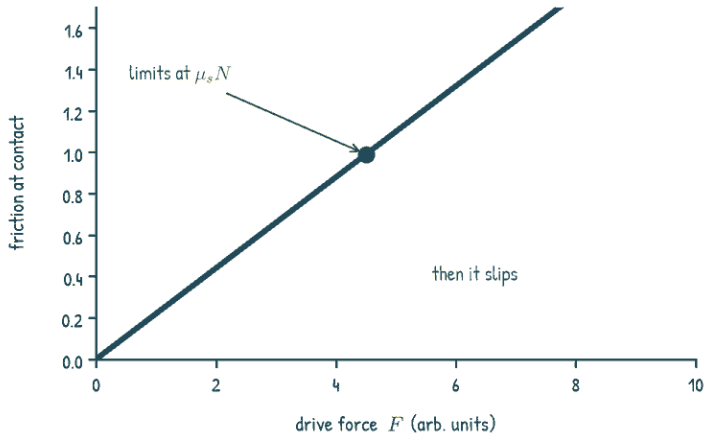
A rolling wheel does not slide, so the friction at the contact is static and the contact point is at rest

the contact point is instantaneously at rest



$$v_{cm} = \omega R$$

rolling uses static friction



KEEP / a rolling wheel uses static friction; the contact point is instantaneously at rest

What are the four moves that solve almost every problem here?

? The chapter has fourteen sections and one page of results. Which four procedures actually carry the load?

1

System, constraint, energy, frame

Every worked problem in this edition is one of those four. Sum forces over a system to remove internal unknowns; differentiate a length constraint to link accelerations; use energy to get speeds without integrating; change to the accelerating frame to turn dynamics into statics.

$$\sum F_{\text{ext}} = Ma_{\text{cm}}, \qquad x_1 + 2x_2 = \text{const}$$

$$\Delta K + \Delta U = W, \qquad \vec{g}_{\text{eff}} = \vec{g} - \vec{a}_0$$

2

Choosing between them

Keyword routing is the fastest method. Pulley words mean constraint. "Find the force between" means system first, then isolate. "Tension at an angle" means energy plus one radial equation. "In the lift or the car" means change frame.

→ if two routes disagree, the error is in the algebra, not in the physics

3

The discipline they share

Each shortcut deletes a category of unknown rather than solving for it. That is why they are shorter, and also why they are dangerous: if the deleted unknown was actually needed, the method silently gives an incomplete answer.

4

A worked routing

Two blocks on a wedge, a string between them over a pulley, the wedge accelerating. Which shortcut first?

$$\text{constraint first (string), then system, then frame}$$

→ three moves, three unknowns; the FBD-only route needs six equations

GATE

routing table

incline → resolve; pulley → differentiate; circular → centre; banked → $\tan\theta$; accelerating → frame; multi-body → system

TRAP

reaching for the shortcut first

Write the free-body diagram anyway, at least in your head. The shortcut is a compression of that diagram, not a replacement for it.

INSIGHT

the two-frame habit

Once a frame is chosen, write it at the top of the page. Most sign errors in this chapter are frames that were mixed mid-solution.

Four shortcuts, one theme: replace a hard dynamics question with a smaller one

1 system equation

$$\sum F_{\text{ext}} = Ma_{\text{cm}}$$

a always, internal forces never

2 differentiate the constraint

length → velocity → acceleration

never guess the ratio

3 energy for the speed

$$v^2(\theta) = v_0^2 - 2gR(1 + \cos\theta)$$

then one radial equation for T

4 one vector of gravity

$$g_{\text{eff}} = \sqrt{g^2 + a_0^2}$$

then solve a static problem

KEEP / four shortcuts, one theme: replace a hard dynamics question with a smaller one

Can a monkey climb away from its own counterweight?

? A monkey and an equal counterweight hang from one rope over a pulley. The monkey climbs. Does the counterweight stay, rise, or fall?

1 ## Write the two equations

Both sides have mass m. The tension is the same on both because the rope is massless and the pulley is ideal. Take upward as positive on each side.

$$T - mg = ma, \qquad T - mg = ma$$

→ identical equations, so the two accelerations are identical

2 ## The climb rate never enters

The monkey climbing means its centre of mass moves relative to the rope. But the rope and the monkey are not the same body: the tension acts on the monkey, and the monkey pulls the rope down with the same force. The rope still has tension T.

$$a = \frac{T}{m} - g \quad \text{on both sides, for any } u$$

3 ## What actually happens

Whatever the monkey does to the rope, it does the opposite to itself, so the pair moves as one system with the common acceleration set by the external weights. If the monkey hauls on the rope, both rise together; if it lets rope slip, both fall together.

→ the monkey can change the tension, but not the ratio a/g = T/mg − 1

4 ## A worked climb

A 6 kg monkey and a 6 kg counterweight hang over a light pulley. The monkey climbs the rope at 0.5 m/s relative to it. Find both accelerations and the tension.

$$a_1 = a_2 \Rightarrow T - mg = ma \text{ on both sides}$$

$$\text{grip without slipping}: \; a = 0, \; T = 58.8 \text{ N}$$

→ whatever the climb rate, both rise at the same speed: only the distance to the pulley changes

KEY RESULT

the whole result

$$T - mg = ma \text{ on each side} \Rightarrow a_1 = a_2$$

the climb rate u cancels identically

NUMBERS

a check

m = 5 kg each, monkey climbs at 1 m/s relative to the rope.

T = 2mg = 98 N? No: T = m(g + a), and a is free.

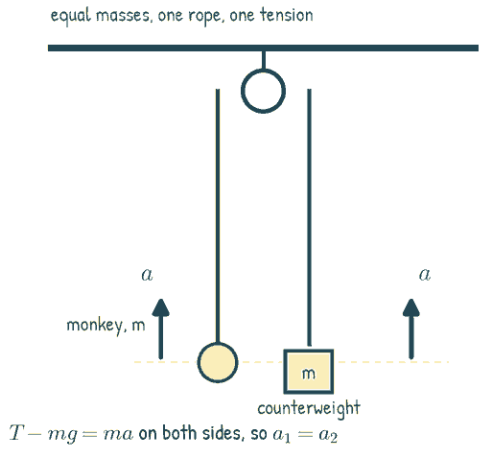
If the monkey grips and does not slip, a = 0 and T = 49 N.

TRAP

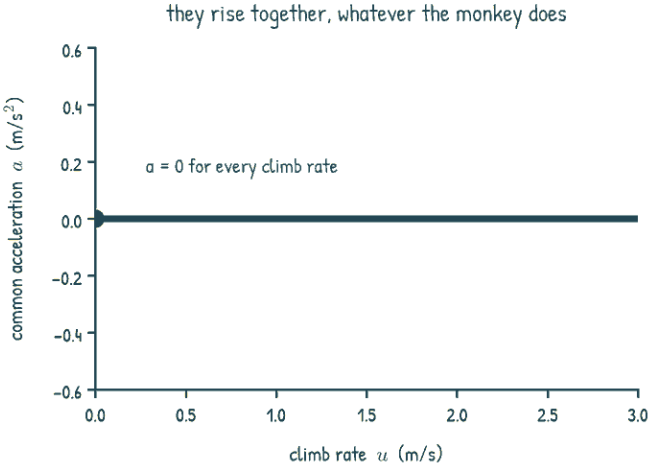
thinking the monkey can win

Equal masses on one ideal rope cannot separate. The monkey rises, the counterweight rises, at the same rate — the monkey only closes the distance to the pulley faster.

Equal masses on one rope accelerate together: the monkey cannot outclimb its own counterweight



$T - mg = ma$ on both sides, so $a_1 = a_2$



KEEP / equal masses on one rope share one acceleration: the climb rate drops out entirely